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Probing Foundation-Model Embeddings for Fetal Congenital Heart Disease Detection

A Controlled Evaluation of Discrimination and Clinical Utility

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Abstract

This dissertation asks whether frozen embeddings from general-purpose vision foundation models can detect congenital heart disease on fetal-cardiac ultrasound video (cine clips) by direct probing of frames sampled from those clips, and whether they are clinically useful at the low prevalence of a real screening population, where the only decision that matters is whether to refer. The two questions are reported separately.

On whether it works, a light classifier trained on frozen embeddings from a general-purpose model reaches an area under the receiver-operating-characteristic curve of 0.825 on the four auditable lesions (transposition of the great arteries, atrioventricular septal defect, tetralogy of Fallot and hypoplastic left-heart syndrome). A domain-pretrained backbone reaches 0.911, an improvement of 0.086 that a scale-matched control attributes to the pretraining domain rather than to model size. At 0.95 specificity the domain backbone catches 0.593 of affected cases, a level a screening programme would find usable.

On whether it is useful, the work leads with sensitivity at a fixed high specificity and prevalence-adjusted predictive values: at a one-per-cent prior a positive call carries a positive predictive value near 0.11, weak as a rule-in, while a negative call is strongly reassuring, so the tool is a rule-out test and sits below a modern national programme at the operating point a service would run.

The lead finding is one of feasibility: a representation learned with no cardiac supervision already encodes enough of the structure that screening requires for direct probing to work, a regime the field has assumed needs domain-specific training from scratch. A general-purpose backbone is pretrained once, at the expense of a well-resourced laboratory, then released and reused frozen, so a downstream task like fetal-cardiac screening avoids both collecting a domain corpus and training a model from scratch. As the community improves these models further, such a task inherits those gains by reusing a stronger backbone rather than re-investing in pretraining.

Acknowledgements

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Introduction

1.1 A decision made under a one-per-cent prior

Congenital heart disease (CHD) is the most common serious birth defect, present in roughly eight of every thousand live births [1]. The cases that decide the outcome are the severe **duct-dependent** lesions, where what happens in the first hours after birth depends on whether the defect was already known. Two of the conditions that this work targets show why (Figure 1.1). In transposition of the great arteries (TGA) and in hypoplastic left-heart syndrome (HLHS) the fetus is stable in the womb, because the placenta does the work of the lungs and the arterial duct holds the circulation open. The danger arrives at birth. As the duct closes during the first days of life, an undetected newborn can collapse without warning, sometimes only after going home, at which point the chance to intervene safely has reduced.

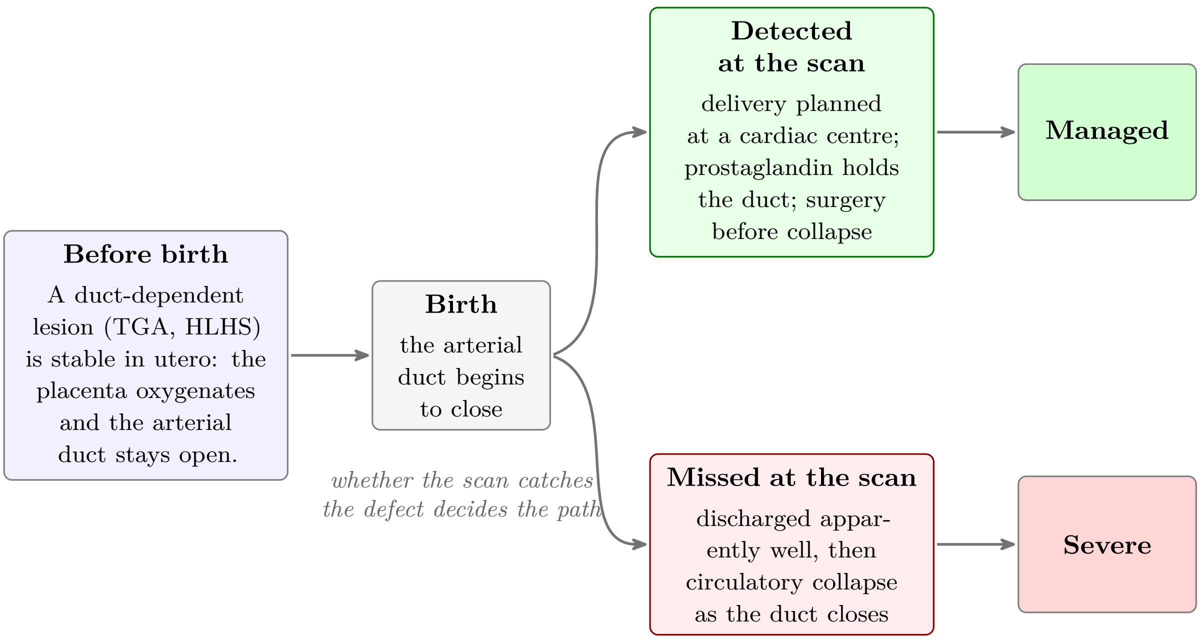

Figure 1.1. The clinical stakes of the anomaly scan for a duct-dependent lesion. The same heart follows two very different courses, decided only by whether the defect was caught before birth.

A defect found before birth changes that course of bad outcomes completely. The delivery is planned at a centre with paediatric cardiology care present there, and the duct can be held open with medication (prostaglandin) until the baby's heart is assessed and a surgery is arranged before the child becomes critically unwell. The main value of the scan is therefore not in naming the lesion, but to catch it. A defect that is caught is for the most part managed. The missed ones are the the ones that become severe.

In the United Kingdom that catch has to happen during the routine mid-trimester anomaly scan, performed between eighteen and twenty-one weeks, when a sonographer examines the fetal heart and decides whether it looks normal enough to pass [2]. It is a difficult examination. The structures are small and moving and the operator has to acquire the image, tune the machine, and read the anatomy at the same time. Plus, almost every heart on the list is normal, which is also why a rare abnormality like CHD is easiest to overlook by human sonographers working tirelessly. The data this project uses is what these scans record: short moving video clips of the beating heart, from which still frames are sampled (not isolated photographs). On average, antenatal scans pick up only around forty-five per cent of CHD cases, and detection varies widely from one service to another [3]. Scan volumes keep rising while expert fetal cardiologists stay scarce, so consistent detection is hard to sustain. That gap is where an automated tool could help. Not for replacing the sonographer, but as a second reader that flags a heart for a closer look.

So what this setting needs follows from the arithmetic of screening. CHD is present in about one per cent of pregnancies, and the two ways of being wrong cost very differently. For the children who most need early management, a missed lesion like the one above is irreversible. A false alarm is mild for one family but costly across a whole programme. At a one-per-cent prior, a test that wrongly flags five per cent of healthy pregnancies raises about five false alarms for every true case, and a national programme referring at that rate would swamp the specialist fetal-echocardiography services the referrals depend on. So a useful tool keeps its false-positive rate very low and recovers as much sensitivity as it can at that setting. That is why this work judges a model not on how cleanly it separates affected from healthy hearts in the abstract, but on whether it would improve the referral decision at an operating point a service could actually run.

1.2 The question this dissertation asks

Most automated work on the fetal heart takes the expensive route. It trains a large model from end to end on scarce, carefully labelled cardiac data (see Section 2.5). There is a cheaper route. It reuses a frozen, already-trained representation and fits only a light classifier on top. Each subject gives one or more video clips. A general-purpose vision foundation model [4], trained on ordinary images with no medical supervision, turns each frame sampled from those clips into an embedding: a numerical feature vector. If that embedding holds enough of the relevant structure, a small classifier fitted on the frozen features may be enough to read a congenital defect from the scan. The model that produced the features stays untouched. Probing a frozen general-purpose representation this way has not been tried on the fetal heart, and there are reasons to doubt it would work. The representation was learned with no exposure to congenital heart disease. Ultrasound looks nothing like the everyday images behind it. And the public fetal-ultrasound data a domain model could be trained on is overwhelmingly non-cardiac.

If it works, the appeal is strategic. A general-purpose backbone is pretrained once by a well-resourced laboratory and released. A downstream task then pays only to fit a light head on its frozen features. It can ride later improvements by swapping in a stronger backbone, and never has to pretrain its own. This work tests the premise that the representations are good enough today. It then asks how best to use them. Which backbone, which head, how much of each scan to sample.

This work tests **representational sufficiency**. The question is not whether the approach is possible. It is whether a representation learned without any cardiac supervision already encodes the structure that screening needs. The work poses it as one question in two parts:

> Can general-purpose foundation-model embeddings represent fetal-cardiac ultrasound well enough to detect congenital heart disease by direct probing? And at the roughly one-per-cent prevalence where the only decision that matters is whether to refer, are they clinically useful, not merely good at ranking?

The two parts are distinct, and the work answers them separately. Whether it works is a question of discrimination: how cleanly the embeddings separate affected hearts from healthy ones. Whether it is useful is a question of the decision. At a false-referral rate a service could absorb, does the model catch enough true cases to be worth running? And what does a positive or negative call mean when only about one pregnancy in a hundred is affected? A model can do the first well and the second badly. Keep them apart, and the evaluation can give a clinician something to act on.

Both parts are settled by controlled comparison, not a single pass-or-fail test, and that comparison holds most of the evidence. One factor changes at a time. The head, the data, the splits and the metrics are all held fixed. Only the backbone changes, so a general-purpose representation can be weighed against a domain-pretrained one on equal terms. The head is then varied on a fixed representation, to ask what turns a given degree of separation into a dependable decision. Varying how much of each scan the head sees separates a limit in the information from a limit in the model. The two questions then become a study of which design choices drive discrimination and which drive the clinical decision.

This is a feasibility result. It is not a finished screening tool, and it claims no more. The evidence is one research cohort, a backbone trained outside the domain, and a light classifier on top.

1.3 Refer, or do not refer

The decision a screening tool informs is binary. It does not matter whether the sonographer can name the exact lesion. What matters is whether the heart is sent for the specialist look it needs. The evaluation is built around that one decision. Its headline task is the binary one: any of the nine congenital conditions in the cohort against a structurally normal heart. A second task is more clinically grounded. It groups the four conditions a national programme measures its own detection against into one referable label: TGA, atrioventricular septal defect (AVSD), tetralogy of Fallot (TOF) and HLHS. This dissertation calls these the **auditable four**. They are the lesions a screening programme audits its performance on, and grouping them mirrors the only decision the setting asks for, to refer this heart or not. Results for individual conditions are reported, but they are read as description. The decision the work cares about is refer or do not refer.

Such a tool earns its place where the scan is hardest to deliver well. In well-resourced programmes antenatal detection of congenital heart disease is already reasonable. It is pooled at around forty-five per cent across unselected populations [3], though it swings widely by region even within one national service, from roughly a third of cases to over four-fifths [5]. In many lower-resource settings it is lower still, with antenatal detection of structural anomalies reported near a sixth of cases in African series and a third in Asian ones [6]. A tool that catches a useful share of cases at a false-referral rate a service can absorb has the most to offer where current detection is lowest, not in competition with the strong average of a mature programme. That is the upside this work speaks to, claimed where it holds and no further.

1.4 Contributions

This work is an evaluation. It does not propose a new model, and its contributions are led by what the evaluation establishes.

1. **Frozen general-purpose embeddings are sufficient for fetal-cardiac CHD detection.** A frozen backbone that has almost certainly never seen a congenital defect, carrying only a light head on its embeddings, separates the auditable lesions from healthy hearts on the iFIND cohort, at an auditable-four AUROC of 0.825. The field has assumed domain-specific training from scratch is required. This is the feasibility result, and the one the work leads with. The claim rests on a controlled comparison across three axes: the representation, the head and the sampling budget, each varied with the rest held fixed. The choice that drives discrimination is not the one that drives the clinical decision. The Discussion works out why. The evaluation, as much as any single model, is the contribution.

2. **Which representation drives discrimination.** With the head held fixed, swapping a general-purpose backbone for a domain-pretrained one lifts auditable-four discrimination by 0.086 AUROC, to 0.911. A scale-matched control attributes that gain to domain pretraining, not model size: the advantage is +0.074 at matched size. The gain is a few points on AUROC, but it is clear at the screening operating point. There, sensitivity at 95% specificity is about 0.59 for the domain-pretrained backbone against below 0.40 for the general-purpose one.

3. **Which head drives the decision.** At the screening operating point the simplest head, a mean-pooling linear probe, is the only head family tested there whose decisions stay reliable on every fold and every backbone. The higher-AUROC transformer head does not, nor does the cross-attention head. Calibration at the threshold is a property of the head, not the representation, and it is what decides the referral. So a domain-specialist backbone alone does not secure the best clinical decision.

4. **What input scale and sampling buy, and where the ceiling is.** Scaling the per-subject token budget to roughly twenty times the baseline lifts both discrimination and screening sensitivity. Discrimination rises a little, by 0.036 on auditable-four AUROC. Sensitivity at 0.95 specificity rises more, by 0.173, about thirty per cent in relative terms. Adding new clips helps more than sampling the frames within a clip more densely, though the effect is directional only. The clinical-decision ceiling holds across the budget all the same. The binding constraint is the weak subject-level supervision, not the head or the sampling. A designed set of falsifications rules out the alternatives: head capacity, training recipe, clip aggregation and cardiac-view-aware selection.

5. **A per-fold estimand for the clinical-utility metric under cross-validation.** The threshold-based utility metric is computed within each held-out fold and summarised across folds, rather than pooled across the concatenated predictions. The two are not interchangeable. Only the per-fold form respects the subject-disjoint partition the rest of the evaluation rests on.

6. **A postnatal-confirmation cohort-cleanup procedure.** A confirmation flag decoded from undocumented filename metadata restricts the coarctation slice to postnatally confirmed positives. Coarctation has the weakest antenatal predictive value. The cleanup lifts its five-fold AUROC from about 0.858 to 0.879, with no

change to any model.

1.5 How this dissertation is organised

The dissertation follows the two-part question. Chapter 2 sets out the screening task and the foundation-model method, and locates the gap in the literature. Chapter 3 specifies the cohort, the controlled comparison and the metrics, with the analysis plan fixed in advance. The evidence is then split along the two questions. Chapter 4 asks whether the embeddings work. It reports discrimination and the sensitivity reached at a usable operating point. Chapter 5 asks whether they are useful. It reports what a referral call means at screening prevalence, the net benefit of acting on the model, and the calibration of its probabilities. Chapter 6 reads these together and sets the limits, and Chapter 7 concludes. The feasibility result is the lead takeaway throughout. Clinical utility and calibration bound that result rather than rival it.

Chapter 2

Background and Related Work

This work puts a clinical screening task next to a method. The method probes a frozen foundation-model representation with a light classifier. Judging whether it fits the task takes some groundwork. The chapter sets out what the fetal cardiac screen asks of an image, what a foundation model is and why its features might carry the structure the screen depends on, and what has already been shown when these models meet medical ultrasound, and fetal ultrasound in particular. It then closes on the gap the work addresses.

A few ideas run through the chapter and come back later. The screen is a rare-event decision, which is why ranking and clinical utility come apart. Probing a frozen backbone is cheap, and that is what makes the approach worth testing at all. The supervision is only a single label per subject, and that is where the work eventually meets its ceiling.

2.1 The fetal cardiac screen as an imaging problem

Screening for congenital heart disease is a problem of image acquisition first, and a classification problem second. The fetal heart is examined live, by hand, through a narrow acoustic window. The sonographer is not there to name the lesion. What matters is whether the heart looks normal enough to pass. Two properties of that task shape everything a model built on top of it can do. The examination follows a small set of standard views, and the images themselves are unusually hard to read.

Cardiac assessment in the mid-trimester anomaly scan is organised around a sequence of standard cross-sectional planes, each positioned to bring particular structures into view [7]. The four-chamber view anchors the examination. The outflow-tract and three-vessel views were added to national protocols because a four-chamber view alone misses conditions where the great arteries are malformed while the chambers still look balanced. Table 2.1 lists the planes and the principal lesions each is placed to reveal, and Figure 2.1 shows a representative de-identified frame from the cohort for each plane. The cohort this work uses spans these planes, though the method samples frames across clips without selecting views by hand. The conditions each plane exposes are why the evaluation later reads the auditable four as one referable group, instead of chasing each subtype on its own.

The second property is that ultrasound is a difficult image to read, for an automated system as much as for a person. The image is formed from the interference of backscattered echoes, which gives it a granular speckle texture, quite unlike the additive noise most vision models are trained against. Dense structures such as the spine or a rib cast acoustic shadows, and a shadow falling across the interventricular septum can imitate a septal defect or hide a real one. Each frame is only a two-dimensional slice, so a small change in

Table 2.1. The standard fetal cardiac screening planes and the principal lesions each is positioned to reveal. The four-chamber view anchors the examination. The outflow-tract and three-vessel views improve detection of conditions a four-chamber view alone can miss.

View	What it shows	Lesions it can reveal
Four-chamber	Ventricular symmetry, atrioventricular valve offset, the crux of the heart	Hypoplastic left-heart syndrome, atrioventricular septal defect, large ventricular septal defects
Left ventricular outflow tract	Continuity of the septum with the aorta	Ventricular septal defect, transposition of the great arteries
Right ventricular outflow tract	The pulmonary artery crossing the aorta	Pulmonary stenosis, transposition of the great arteries
Three-vessel and trachea	Alignment and calibre of the great vessels and the aortic arch	Coarctation of the aorta, right aortic arch

probe angle changes how a structure appears. A normal heart can look abnormal from the wrong plane, and a defect can be lost behind a convenient one. The heart is also moving. At a fetal rate of 120 to 160 beats per minute a full cycle lasts between 375 and 500 milliseconds, the shorter cycle at the faster rate, around eleven to fifteen frames at a typical 30 fps acquisition rate. Some abnormalities show only in that motion, which is why the examination is captured and modelled as short video clips, not single still images. A representation built for everyday photographs has seen none of this. That is the first reason to doubt it would transfer, and the reason this chapter returns to the domain gap in Section 2.5.

2.2 Evaluating a screening test

The rare-event setting of screening makes some standard measures misleading and others essential, so how a model's output is judged matters as much as how it is produced. A clinical prediction model is read along three axes. **Discrimination** is how well it ranks affected hearts above healthy ones. **Calibration** is whether its probabilities can be trusted as numbers. **Clinical utility** asks a different thing: does acting on the model improve the decision? The two-part question this dissertation asks, whether the embeddings work and whether they are useful, maps onto the first axis and the third, and calibration is the bridge between them. The measures for each axis are defined here, so the results chapters can lean on them rather than introduce a metric and a finding in the same breath.

Sensitivity, specificity and the operating point. A score becomes a decision only at a threshold, and the threshold sets the trade-off between the two ways of being wrong. Sensitivity is the fraction of affected cases a test flags. Specificity is the fraction of healthy cases it correctly passes. In a screening programme the false-referral rate is what bounds the system. Almost every scan is normal, so specificity is held high, near 0.99 in a mature programme, and the question becomes how much sensitivity is available there. This work therefore leads with sensitivity at a fixed high specificity. A summary averaged over every threshold includes the permissive settings a programme would never use, so it is the wrong headline here.

Predictive values depend on prevalence. A clinician works from the predictive values. Given a positive call, how likely is the fetus to be truly affected? That is the positive

(a) Four-chamber view

(b) Left ventricular outflow tract

(c) Right ventricular outflow tract

(d) Three-vessel view

Figure 2.1. Representative de-identified frames from the iFIND cohort, one for each of the four standard screening planes (compare Table 2.1), each sampled from a video clip. The plane assignments shown are those of an automated fetal standard-plane classifier applied to structurally normal hearts, given here as representative examples rather than clinically certified labels. Each frame is cropped to the ultrasound sector and is the kind of image the frozen backbone receives.

predictive value. Given a negative call, how likely is it to be healthy? That is the negative predictive value. Both depend on how common the condition is. A research cohort enriched with disease inflates the positive predictive value relative to a screening population, so a value measured on the cohort has to be transported to the deployment prevalence before it means anything operationally. Given the test's sensitivity and specificity and a target prevalence, the predictive values follow from Bayes' rule. At a prevalence near one per cent, a positive call can be far weaker than the same test's cohort figure suggests.

Calibration grades the probabilities. A model can rank cases well and still output probabilities that are systematically too high or too low. That matters because the referral acts on the probability at a threshold. Calibration is read by the **Brier score**, the mean squared error of the predicted probabilities. The Brier score splits into three parts: a reliability term (how far the probabilities sit from the observed frequencies), a resolution term (how much they vary with outcome), and an irreducible uncertainty term set by the base rate. It is read too by the **expected calibration error**, the average gap between confidence and accuracy across bins of the score. A strong ranker with poor calibration can decide worse than a weaker ranker that is better calibrated, and this work meets that case in practice.

Net benefit weighs the decision. Discrimination and calibration describe the scores. Clinical utility asks whether acting on them helps. **Decision-curve analysis** compares a model against the two trivial policies, referring everyone and referring no one, across a range of decision thresholds [8]. The net benefit at a threshold probability p_t counts true referrals against false ones, weighted so that the threshold encodes a cost ratio. To act at

p_t is to treat a missed case as $(1 - p_t)/p_t$ times as costly as an unnecessary referral. At a threshold of 0.20 that ratio is $(1 - 0.20)/0.20 = 4$, so acting there judges a missed defect four times as bad as a false alarm. The framework makes that value judgement explicit. A model is worth acting on at a given threshold only if its net benefit clears both trivial policies.

The limitation of the area under the ROC curve. The most reported discrimination summary is the area under the receiver-operating-characteristic curve, the probability that a random affected case is ranked above a random healthy one. It is convenient and threshold-free, but it averages over every operating point, including the permissive ones a programme would never adopt. Screening runs under extreme class imbalance, near 0.99 specificity, so the operating point that matters lies far from where most of the curve's mass sits. A strong area under the curve need not imply a useful test. A precision-recall summary, which reflects performance on the rare positive class, can be more revealing. The measure is reported throughout because it is the field's common currency, but it is read as a check, not the headline.

The detection landscape. A model's numbers mean little without the practice it would join, and antenatal detection of congenital heart disease varies widely today (Figure 2.2). A national programme sets itself a minimum standard near fifty per cent [2]. Pooled detection across unselected populations is around forty-five per cent [3]. A mature programme's per-condition audit reaches much higher, the upper reference for what is achievable [9]. An inexpensive automated aid has the most to add where detection is currently lowest, and the results return to that niche. The formal definitions and estimands behind these measures are given in Section 3.7.

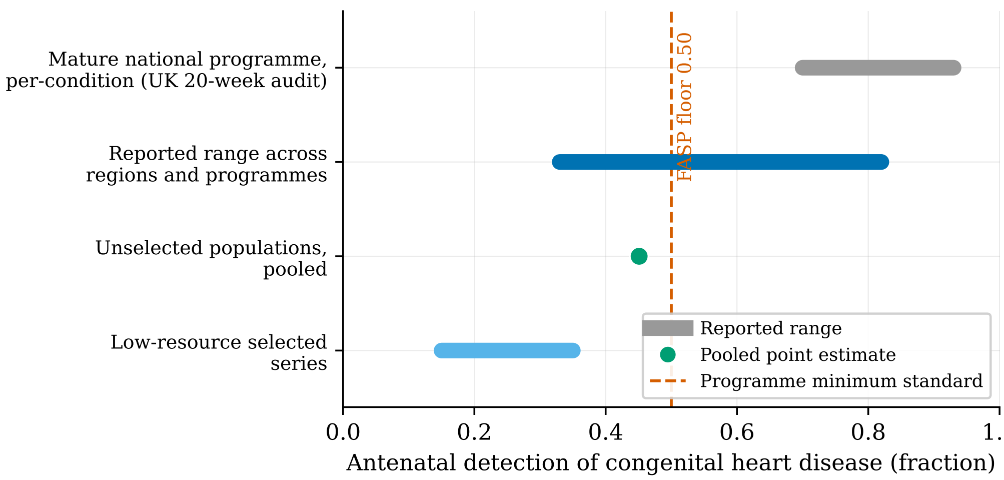

Figure 2.2. The landscape of antenatal congenital heart disease detection across care settings. Reported detection ranges from roughly 0.15 to 0.35 in low-resource selected series [6], through a pooled value near 0.45 in unselected populations [3], to 0.70 to 0.93 per condition in a mature national programme, the United Kingdom twenty-week audit [9]. Rates reported across regions and programmes span roughly 0.33 to 0.82 [5]. The dashed line marks the 0.50 programme minimum standard [2]. The model is not shown here. Its per-lesion performance against the same references is in Figure 5.1 (Section 5.1).

2.3 Foundation models and frozen representations

The method this work evaluates rests on a recent shift in computer vision. A **foundation model** is a large network pretrained on a very large, unlabelled image collection, using an objective that supplies its own supervision. The result is a general-purpose encoder

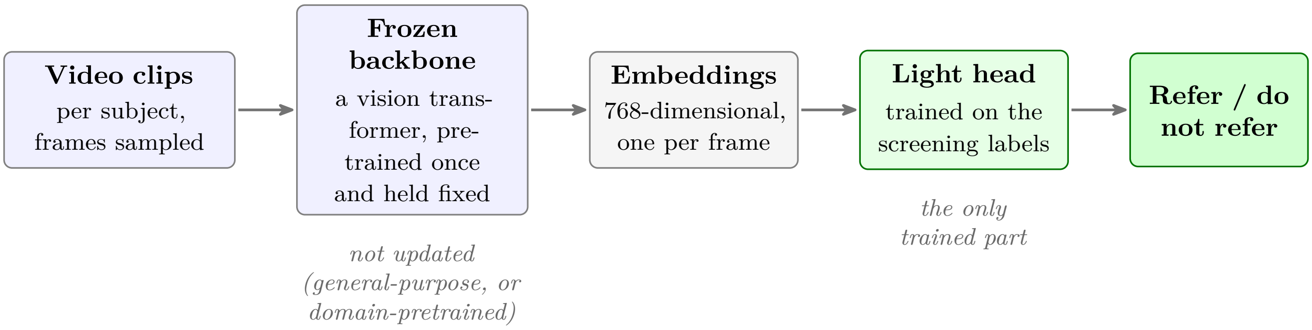

Figure 2.3. Probing a frozen foundation model. The backbone, pretrained once and held fixed, turns each frame sampled from a subject's video clips into an embedding. Only a small head is trained, on the screening labels, to turn those embeddings into a referral decision. The same head can sit on a general-purpose backbone (DINOv2) or a domain-pretrained one (FetalCLIP), so the representation can be changed with everything else held equal.

whose features transfer to many downstream tasks with little or no further training [10]. The parallel with the move to pretrained language models is deliberate, and the practical appeal for medical imaging is the same. Expert annotation is scarce and expensive. A strong pretrained representation can be reused instead, so a small classifier fitted on its frozen features can stand in for a model trained end to end from scratch. Figure 2.3 shows the overall shape. The heavy backbone is pretrained once and held fixed, and only a small head is trained on the screening labels, on either a general-purpose or a domain-pretrained representation.

2.3.1 Self-supervised learning and the vision transformer

Two families of self-supervised objective produce such representations. **Contrastive** methods pull together different augmented views of the same image and push apart views of different images, so that semantically similar images sit close together in the feature space. SimCLR and MoCo are the standard examples [11, 12]. **Self-distillation** methods instead train a student network to reproduce the output of a teacher across different views of the same image, where the teacher is a slowly moving average of the student. This avoids the representational collapse that contrastive methods hold off with large batches of negative samples. BYOL introduced the approach, and DINO carried it to vision transformers [13, 14].

Both backbones used in this work are vision transformers. The architecture takes an image, divides it into a grid of fixed-size patches, and projects each patch to a vector. A learned position embedding is added so the model knows where each patch sat, and a single extra token, the class token, is prepended to summarise the whole image. This sequence passes through a stack of transformer encoder blocks, in which every token attends to every other, and the output at the class token is taken as the image's embedding [15, 16]. Given enough pretraining data the design matches or beats convolutional networks, and it makes no built-in assumption that nearby pixels matter more than distant ones.

2.3.2 DINOv2 and image-text pretraining

DINO, self-distillation with no labels, applies the student-teacher recipe above to a vision transformer [14]. Each image is shown to the student as several crops, some global and some local, while the teacher sees only the global crops, and the student learns to match the teacher across them. The teacher is updated as an exponential moving average of the student, and a centring and sharpening step on the teacher's output keeps the two networks from collapsing onto a trivial solution. The attention maps of a DINO-trained transformer pick out object boundaries with no segmentation labels anywhere in training, which suggests the features carry genuine semantic structure rather than surface statistics.

DINOv2 scales this recipe up in both data and model size, and is the general-purpose backbone evaluated here [4]. Its training set, LVD-142M, is assembled by an automatic pipeline that filters 1.2 billion web images down to 142 million, retrieving images close to a set of curated seeds and removing near-duplicates. Its objective combines the image-level DINO loss with a patch-level masked-prediction loss reused from iBOT, where the student must recover the teacher's representation of masked patches, together with a regulariser that spreads features out in the embedding space [17]. The models come in a family of sizes (Table 2.2), with the smaller ones distilled from the largest. On a frozen-feature linear probe of ImageNet the largest model reaches 86.5% top-1 accuracy, and it is more robust than earlier self-supervised models on out-of-distribution test sets. A frozen representation that transfers this well, with only a linear layer trained on top, is a candidate for a domain it never saw during pretraining. The space can be inspected by projecting it to two dimensions with a neighbour-embedding method such as UMAP and scoring how cleanly the classes separate with a silhouette coefficient. A representation that shows no separation in two dimensions can still carry the structure a classifier needs in its full dimensionality, and the discussion returns to that distinction.

Table 2.2. The DINOv2 model family. The embedding dimension is the width of the vector a downstream head receives for each image. The smaller models are distilled from the ViT-g teacher. This work uses the 768-dimensional ViT-B/14 model as its general-purpose backbone.

Model	**Parameters**	**Embedding dimension**
ViT-S/14	21M	384
ViT-B/14	86M	768
ViT-L/14	300M	1024
ViT-g/14	1.1B	1536

DINOv2 learns from images alone. A second route to a general-purpose encoder learns from images paired with text. CLIP trains an image encoder and a text encoder together, on around 400 million image-caption pairs, so that an image and its caption land close in a shared space while mismatched pairs are pushed apart [18]. The image encoder that results is a strong general-purpose feature extractor, and because it shares a space with language it can also classify by comparison with text prompts. FetalCLIP applies exactly this recipe within one domain. It is a CLIP-style model trained on 210,035 fetal ultrasound images paired with captions, with a vision-transformer image tower, and it is the domain-pretrained backbone evaluated here [19]. It learned on a large fetal-ultrasound corpus, but its training target was to match images to captions, not to recognise specific heart defects. So it carries domain familiarity without disease supervision. Both backbones expose a 768-dimensional image embedding, which lets the later comparison hold the classifier fixed and change only the representation.

2.4 From frame embeddings to a subject decision

A frozen backbone turns each frame into an embedding, but a screening decision is made for a subject, from many frames across several clips, so something has to combine them. The options run from fixed pooling to learned temporal models. A set of per-frame embeddings can simply be averaged. A sequence can be passed to a recurrent or three-dimensional convolutional model. At the expressive end, it can go to a video transformer that attends across space and time, as in TimeSformer and ViViT [20, 21]. The cost rises along that range, in computation and in the labelled data needed to fit the extra parameters. This work sits towards the cheap end, pooling frozen embeddings under a light head instead of training a bespoke video model.

The pooling choice is not neutral. Mean pooling is simple and stable, but it discards variation across frames. The embedding of a frame in systole differs from one in diastole, and averaging collapses that difference into a single static representation. Max pooling keeps the strongest response in each dimension, but it is sensitive to a single noisy frame and has no clear meaning in a learned feature space. Attention pooling learns a weight for each frame, so the head can lean on the informative frames and discount the rest. This is the mechanism of attention-based multiple-instance learning, where a bag of instances carries one label [22]. A small transformer over the frame embeddings is the most expressive option. It can model relationships between cardiac phases, at the price of more parameters and a greater appetite for data.

For the heart, the choice matters because of motion. A cardiac cycle lasts only a few hundred milliseconds, and some abnormalities show in the movement itself, a restricted valve or an asynchronous contraction, not in any single still frame. Mean pooling over a full cycle converges towards a motionless average that discards exactly that signal. Whether a richer aggregation recovers enough of it to matter is an empirical question. So is whether more frames or more clips is the better way to spend a fixed token budget. This work examines both directly when it varies the head and the sampling.

2.5 Foundation models in medical imaging, and the gap

Whether features learned from everyday photographs survive the move to medical images is not obvious. Medical images have different contrast and texture, and their structures of interest, a tumour or a malformed valve, look nothing like the cars and faces that dominate natural-image collections. The evidence so far is nonetheless encouraging. Frozen foundation features give strong linear-probe results across magnetic resonance, computed tomography, radiography and dermoscopy, and self-supervised pretraining has been shown to improve both data efficiency and robustness for diagnostic imaging [23–26]. The usual explanation is that these models learn low-level primitives and a separation of object from background that remain useful across domains. Whether that holds for ultrasound, with its speckle and its strong view dependence, is an empirical question, and it is the one this work puts to the fetal heart.

Several recent works circle that question without answering it. Table 2.3 sets them beside the approach taken here, and the prose adds what the table cannot, which is why each one matters. Two of them establish the premise this work builds on. Schulthess and Konukoglu fit a density model over frozen DINOv2 embeddings of healthy images and score anomalies by deviation from it. Their telling result is that the frozen features stay aligned with anatomy even in abnormal images, direct evidence that a representation trained only on natural images still carries medical structure [27]. L-FUSION, from the supervisor's own research group, is the closest methodological precedent. It trains light heads on a frozen ultrasound encoder and proves that a foundation model which preserves the information a task needs yields embeddings that are a sufficient statistic for both the prediction and its uncertainty [28]. That theorem is the footing for probing a frozen encoder rather than fine-tuning one, the design this work adopts.

The other two mark the boundaries the present work has to clear. MM-DINOv2 shows that adapting a foundation model to a medical domain works, but it fine-tunes the backbone, and its frozen variant was the weakest model it compared [29]. So frozen probing cannot be assumed to succeed, and has to be demonstrated. STUD is the most direct prior work of all, the only one on the same clinical target. It detects congenital heart disease in fetal-cardiac video, but by modelling normal hearts and flagging any deviation, so it never trains on disease and never asks how its detector would behave as a referral test at screening prevalence [30]. Each work comes at the question from a different side. None takes a frozen, general-purpose representation and asks directly whether it already suffices.

Table 2.3. Recent work closest to this evaluation, and what does not carry over to probing a frozen general-purpose representation on the fetal heart.

Work	Approach	Domain / task	What does not carry over
MM-DINOv2 [29]	Fine-tunes a multi-modal DINOv2	Brain-tumour subtyping, multi-sequence MRI	Adapts the backbone rather than freezing it. Its frozen variant was weakest. Multi-sequence MRI, not single-stream ultrasound
DINOv2 anomaly detection [27]	Density model over frozen DINOv2 embeddings	Unsupervised anomaly detection, brain and retina	Unsupervised normality scoring, not a supervised referral. A precision gap on subtle anomalies
L-FUSION [28]	Trainable heads on a frozen ultrasound encoder	Fetal-cardiac (4CH) and head-circumference segmentation, AVSD (CHD) out-of-distribution	A domain-pretrained ultrasound encoder, and segmentation, not classification. Same frozen-encoder principle
STUD [30]	Video self-distillation, normality modelling	Zero-shot congenital heart disease, fetal-cardiac video	Learns only normal hearts and never uses disease labels. A video model. Does not weigh the referral decision at prevalence

Set together, these works leave a specific gap. Adapting a foundation model to the medical domain has been tried, and frozen general-purpose embeddings have been shown to carry medical structure. Training light heads on a frozen ultrasound encoder has both a precedent and a theoretical justification, and zero-shot detection on the exact clinical target exists. The one thing none of them does sits in the middle:

> no published work has asked whether a frozen, general-purpose visual representation, learned with no medical supervision at all, already encodes the fetal heart well enough that a light probe on its embeddings can both detect congenital heart disease and inform the referral decision at screening prevalence.

That is the gap this work addresses, and it is one of representational sufficiency. The work does not propose a new model. It sits at the cheapest end of the range that runs from zero-shot normality modelling to fully supervised training from scratch.

Chapter 3

Methods and Evaluation Design

This work introduces no new backbone. The backbones are taken as published and kept frozen. The heads are small and standard, and the training is ordinary supervised learning on the embeddings they produce. What the work contributes is the evaluation. It is a controlled comparison that holds everything fixed but one factor at a time, so a change in performance can be read back to the choice that caused it. This follows the convention of a clinical evaluation paper, where the evaluation is described alongside the method. What is under study is how existing representations behave on a screening task.

The comparison runs along three axes: which representation produces the embeddings, which head turns them into a decision, and how much of each scan the head is given to see.

The chapter follows that order. It sets out the controlled comparison and the ledger of specified comparisons (Section 3.1), the cohort and the tasks (Section 3.2), and then the three axes as concrete artefacts: the two representations and the embedding pipeline (Section 3.3), the probing heads (Section 3.4) and the per-subject sampling budget (Section 3.5). It fixes the evaluation protocol and the single estimator used throughout (Section 3.6), defines each metric and says why it is used here (Section 3.7), and closes with a reproducibility statement (Section 3.8) precise enough to re-derive every number that follows.

The abstract shape of the pipeline was described in Chapter 2: a frozen backbone turns each frame into an embedding (Figure 2.3), and a light head pools those embeddings into a referral decision (Section 2.4). This chapter instantiates that shape with the specific checkpoints, heads, splits and metrics the evaluation uses, and does not redraw it.

3.1 The controlled comparison design

The pipeline of Chapter 2 has several moving parts: the representation, the head that pools its embeddings, the amount of each scan the head sees, the training recipe and the splits. Three of these are the axes of the study.

The first axis is the **representation**. With the head, training recipe, splits, metrics, and 768-dimensional output width all fixed, only the frozen backbone is swapped. This isolates the impact of a general-purpose foundation model versus a domain-pretrained one on equal terms. The interface remains matched even though the two backbones differ in parameter size, a distinction returned to below.

The second axis is the **head**. On a fixed representation, the function that pools a subject's frame embeddings into a single decision is varied across a family ranging from a plain linear classifier on the averaged embedding to small sequence models that attend across frames.

Holding the representation constant isolates what the head contributes: how much of a usable decision comes from the embeddings themselves, and how much from the machinery placed on top.

The third axis is the **sampling budget**, how much of each scan the head is allowed to see. A subject contributes many frames across several clips, and the number sampled, both within a clip and across clips, sets a per-subject token budget. Varying that budget with the backbone and head fixed asks whether the screening decision is limited by the information presented to the head, or by the head and representation themselves. The budget is defined precisely in Section 3.5.

One objection follows immediately from the first axis. The domain-pretrained backbone (FetalCLIP) is the larger of the two, so an advantage in its favour could be an advantage of size rather than of domain. (The FetalCLIP backbone is open_clip ViT-L/14 that has 300M parameters and a 1024-wide transformer, but CLIP's visual projection head maps that to a 768-d shared CLIP latent space. We store and use the projected output, not the 1024-d hidden state.)

The design answers this with a **scale-matched comparison**. A third backbone is added, a general-purpose model the same size as FetalCLIP (both ViT-L/14, 300M parameters). That gives a ladder of three: the general-purpose backbone at its base size (768-d), the same model scaled up to match, and the domain-pretrained backbone. The scaled-up general model emits a wider embedding, 1024-d rather than 768. As with the FetalCLIP projection above, this is a difference of interface, not of capacity. The head's linear input layer maps either width onto the same hidden dimension, so its capacity and recipe do not change. What the control matches is the backbone size. Any gap that remains between the scaled-up general model and the domain model then separates domain pretraining from scale. The result is reported with the discrimination results.

Beyond the three axes, several specific mechanisms might plausibly raise the screening decision further, and each was specified in advance as a comparison with a stated question and a fixed point of reference. Table 3.1 lists them. Each holds the anchor pipeline fixed and changes only the named mechanism, so that a mechanism which fails to move the decision is itself informative: it rules out that mechanism as the binding constraint rather than leaving it untested.

Two clarifications. The anchor pipeline predicts one referral from a lumped label over uniformly sampled frames, with no hand-selected cardiac views (Chapter 2). Rows that name view selection or per-condition heads test alternatives to that anchor, not steps inside it.

3.2 Data, labels and tasks

3.2.1 The cohort and its labels

The evaluation uses the iFIND research cohort: 4,128 subjects of fetal cardiac ultrasound, each with one or more video clips. Labels are at the level of the subject, not the frame or the clip, and they are taken as given. They come from a curated file provided by the clinical collaboration, in which the original 43 detailed diagnoses have already been collapsed into nine clinical groups, a mapping external to this work and not re-derived here. The nine conditions are transposition of the great arteries (TGA), atrioventricular septal defect (AVSD), tetralogy of Fallot (TOF), hypoplastic left-heart syndrome (HLHS), right aortic arch (RAA), coarctation of the aorta (CoA), pulmonary atresia (PuA), aortic stenosis (AoS) and pulmonary stenosis (PuS). A subject is treated as affected if any of the nine

Table 3.1. The comparisons specified in advance. Within each row the anchor's choice is named first and the alternative it is tested against second. The anchor pipeline is FetalCLIP with the transformer head, at the baseline per-subject token budget, under the subject-disjoint five-fold protocol. A mechanism that does not move the screening decision is still informative, because it rules that mechanism out as the binding constraint. Outcomes are reported in the chapter named in the final column.

Mechanism varied	What it isolates (all else fixed)	Question specified in advance	Reported in
Clip-to-subject aggregation	Flat per-clip pooling against an explicit subject-level hierarchy	Does an explicit hierarchy beat flat pooling?	Chapter 4
Multiple-instance aggregation	The transformer head against gated and ungated attention-based multiple-instance heads	Is the aggregator, not the head, the constraint?	Chapter 4
View-aware sampling	All sampled frames against a top-K cardiac-view relevance filter	Does restricting the head to on-view frames help?	Chapter 4
Motion and dense sampling	Uniformly spaced frames against dense multi-window bursts capturing local motion	Does dense local motion add signal over spread sampling?	Chapter 4
Training recipe	The fixed recipe against a fifteen-configuration search over learning rate, dropout, weight decay, batch size and schedule	Is the shared recipe near-optimal, so fixing one is justified?	Chapter 4
Head capacity	The base head against six variants spanning depth, width, head count, attention pooling and regularisation. Head depth co-varies with the per-subject token budget, so this gives no clean discrimination reading.	Does extra head capacity change discrimination or the decision?	Chapters 4, 5
Per-condition formulation	The lumped binary head with per-condition slicing against a multi-label per-condition head	Do direct per-condition heads beat slicing the lumped head?	Chapter 4
Training prevalence	Class-balanced training against training at the task's own prevalence	Is the clinical-utility gap a train-deploy prevalence artefact?	Chapter 5

labels is positive and as a structurally normal control if all nine are absent.

3.2.2 Two preparation steps

Both act on the clips before any embedding is computed and are applied identically to every backbone. First, clips carrying a colour-Doppler overlay are excluded by a rule-based check on the colour channels, because the coloured flow map is an acquisition choice of the operator rather than anatomy and is not present uniformly across the cohort.

The same Doppler-filtered set of clips feeds every backbone, so the comparison between representations is not confounded by which clips each one saw. Second, the coarctation slice is restricted to postnatally confirmed cases. Coarctation has the weakest antenatal predictive value of the nine conditions, and a postnatal-confirmation flag was recovered by reverse-engineering an undocumented token in the clip-filename convention. Restricting the slice to the confirmed cases is a curation step whose effect is reported in Chapter 4. The standard imaging plane a clip belongs to is available as metadata from an automated plane classifier, but it is not used to select frames. The pipeline samples frames across clips without choosing views by hand, as noted in Chapter 2.

3.2.3 Three tasks at three prevalences

The evaluation is run at three levels of granularity, set out in Table 3.2. The headline task is the binary screening decision: any of the nine conditions against a structurally normal heart. Nested inside it is the primary clinical task, the auditable four introduced in Section 1.3: transposition of the great arteries, atrioventricular septal defect, tetralogy of Fallot and hypoplastic left-heart syndrome, grouped into one referable label against healthy controls, because these are the lesions a national programme audits its detection against and grouping them mirrors the refer-or-do-not-refer decision. The third level reports each condition on its own against healthy controls. The per-condition positive counts range from 27 to 128 across the cohort, only a handful in each fold for the rarest, so these estimates are read as description and not as headline performance. That limit is fixed by the number of affected subjects, and no amount of data sampling can change it.

The two prevalences in Table 3.2 are kept distinct throughout. The cohort prevalence is the proportion affected in this research collection, which over-represents disease relative to the population a screening programme sees and is therefore never read as a deployment figure. The deployment prior is the population birth prevalence: congenital heart disease as a whole is present in about one per cent of births [1, 31], while the four auditable lesions together account for closer to fifteen in ten thousand [32]. The gap between these two priors is wide, and because the predictive value of a positive call depends on the prior, that gap is carried explicitly into the clinical-utility metrics in Section 3.7 rather than left implicit. On the single-split test set defined in Section 3.6, the binary task has 96 affected and 524 healthy subjects, and the auditable four has 62 affected against the same 524.

Table 3.2. The three evaluation tasks. Cohort prevalence is the proportion affected in the research cohort, an enrichment artefact that is never read as a deployment figure. The deployment prior is the population birth prevalence of the relevant lesions. The positive and negative counts on the single-split test set are given in the text.

Task	Positive class	Negative class	Cohort prevalence	Deployment prior
All-nine binary	Any of the nine conditions	Structurally normal heart	0.155	≈1%
Auditable four	TGA, AVSD, TOF or HLHS	Healthy controls only	0.104	≈0.15%
Per condition	A single condition	Healthy controls only	27–128 positives	per lesion

3.3 The two representations and the embedding pipeline

Two frozen backbones. The representation axis weighs the two backbones introduced in Chapter 2. The general-purpose backbone is DINOv2 at the ViT-B/14 size, around 86 million parameters, which emits a 768-dimensional embedding for each frame from its class token. The domain-pretrained backbone is the FetalCLIP image tower, a larger ViT-L/14 model of around 300 million parameters whose output is projected to a 768-dimensional embedding. It was pretrained on 210,035 fetal-ultrasound images paired with captions, so it carries domain familiarity, but it was never trained on heart-defect labels. Neither backbone is updated. In every pipeline it is run once to produce embeddings, and only the head that follows is trained. The encoder is held frozen by design. Frozen probing is the thesis here, the cheap-reuse regime this work sets out to test, so leaving the backbone untouched is the deliberate centre of the design.

A third backbone for the scale-matched comparison. The scale-matched control of Section 3.1 adds one further backbone, a larger general-purpose model, DINOv2 at the ViT-L/14 size, chosen to match the FetalCLIP tower in scale at around 300 million parameters. Its native embedding is wider than 768 dimensions. The head's input projection absorbs that difference, so the rest of the pipeline is unchanged and the comparison turns on backbone scale and pretraining alone. This larger general-purpose model is used only as the size control, and is kept distinct from the ViT-B/14 model that serves as the general-purpose representation in every other comparison.

From clip to embedding. The pipeline that produces the embeddings is the one sketched in Figure 2.3, here made concrete. A subject contributes one or more video clips. From each clip a fixed number of frames is sampled at uniform spacing along its length, and each sampled frame is passed once through the frozen backbone to give its 768-dimensional embedding. A subject is therefore represented by a collection of per-clip arrays of frame embeddings, with the number of clips capped per subject and a seeded random choice made when a subject has more clips than the cap. The same sampling and the same cap apply for every backbone, so a difference between two pipelines cannot come from one backbone having been shown more or different frames than another. How many frames and clips are sampled is itself a variable, the sampling budget of Section 3.5. Run across the cohort under each of the three backbones, the extraction produces cached frozen embeddings for 79,664 clips from 4,128 subjects, the representation on which every head in this work is trained.

3.4 The probing heads and training

3.4.1 A family of heads, from the simplest upward

The head turns a subject's frame embeddings into one referral score, and four are compared, instantiating the pooling families of Section 2.4. The simplest is the mean-pooling linear probe: the frame embeddings of a subject are averaged into a single vector and a balanced logistic regression is fitted on it. It carries no temporal structure at all and serves as the floor the others must beat. The transformer head projects each frame embedding, prepends a learnable summary token, adds a position encoding and passes the sequence through a shallow transformer encoder, reading the decision from the summary token. Its lead configuration is small, two layers of four attention heads over a sixty-four-dimensional hidden state. The cross-attention head is lighter still, a single learnable query that attends across the frame sequence to pool it. The temporal-convolution head applies a stack of dilated one-dimensional convolutions along the frame axis followed by a masked average. The four span the range from a static average to models that can read across cardiac phases. That spread is the point. Mean pooling over a full cardiac cycle averages away the difference between systole and diastole, and the sequence heads are there to test whether modelling that motion changes the decision, rather than to assume it does.

3.4.2 One training recipe, fixed in advance

Every head is trained under the same recipe, so that a difference between heads is a difference of architecture and not of tuning. The recipe uses the AdamW optimiser with a weight decay of 10^{-4}, a learning rate of 10^{-3} (reduced to between 3×10^{-4} and 5×10^{-4} for the attention-based heads), a class-weighted binary cross-entropy loss that upweights the rare positive class, cosine annealing over eighty epochs with early stopping on validation discrimination at a patience of twelve, a batch size of sixty-four, gradient clipping at unit norm, and the final bias initialised to the prior log-odds so that training starts from the base rate. The recipe was fixed after the search recorded in Table 3.1, not chosen by hand, which is what lets a single recipe be held constant across the comparison. All randomness is seeded.

3.4.3 One score per subject

The evaluation decides for a subject, so each head produces one score per subject. The sequence heads score each clip and average the per-clip probabilities. The linear probe averages the embeddings first and scores once. Either way the unit of analysis is the subject, which is why every confidence interval in the chapters that follow resamples subjects rather than clips or frames, and a subject with many clips counts once.

3.5 The per-subject sampling budget

A token, and a budget. The third axis varies how much of each scan the head is given. A token here is one per-frame embedding, the unit the head consumes, and a subject's budget is the number of frames sampled from each clip multiplied by the number of clips kept. The baseline budget samples ten frames from each of up to twenty clips, two hundred tokens per subject, and the budget is raised from there along a grid out to sixty frames from each of sixty clips, roughly twenty times the baseline. The backbone and head are held fixed while the budget changes, so any movement isolates how far the screening decision is limited by the information given to the head rather than by the head or the representation.

Frames against clips. A fixed budget can be spent on more frames within each clip or on more clips, and the two are not equivalent. A fetal cardiac cycle lasts roughly 375 to 500 milliseconds, about eleven to fifteen frames at a typical acquisition rate (Chapter 2), so sampling more frames within a clip buys a denser view of one cycle, while sampling more clips buys more views of the heart across the examination. The grid is laid out to separate these two directions. The budget is set up here as a gradient to be characterised. How the screening decision moves along it is reported with the results. A separate dense, contiguous multi-window sampling appears only as the motion comparison in Table 3.1, not as part of the canonical uniformly-spaced pipeline.

3.6 Evaluation protocol and the canonical estimand

3.6.1 Two evaluation protocols, both split by subject

Every split is made at the level of the subject, so no subject's clips appear in both training and evaluation. The canonical protocol is subject-disjoint five-fold cross-validation: subjects are assigned to five folds by their identifier under a fixed seed, each fold serves in turn as the test set while a tenth of the remaining subjects is used for validation and the rest for training. A second protocol predates it, a single stratified split of the subjects into roughly seventy per cent training, fifteen per cent validation and fifteen per cent test, stratified on the binary label under the same seed. Numbers computed on the 620-subject test set of that single split are reported only as a single-split cross-check and always labelled as such. The cross-validated mean is the headline.

3.6.2 One estimand for every metric

A metric can be computed in more than one way under cross-validation, and the ways do not agree. Throughout this work the canonical estimand is the per-fold five-fold mean, computed by evaluating the metric within each held-out fold and reporting the mean and standard deviation across the five folds. It is reported in preference to two alternatives, pooling the metric over the concatenated predictions of all folds, or using a single split. Pooling a threshold-based metric across concatenated folds would mix subjects the partition deliberately separates. The same per-fold rule therefore governs every metric, the threshold-based clinical-utility ones as much as the discrimination ones. Neither a point estimate pooled across the concatenated predictions of all folds nor a figure from the single split is used as a headline number. The estimators do diverge in practice. The auditable-four AUROC, for instance, is 0.911 as a per-fold five-fold mean but 0.902 on the single stratified split, so which estimator is reported is not a cosmetic choice.

3.6.3 Two bootstrap procedures

Uncertainty is quantified by resampling subjects. A confidence interval on a single metric is a bias-corrected and accelerated bootstrap over subjects, with one thousand resamples. A comparison between two models instead uses a subject-clustered paired bootstrap with two thousand resamples and, within each pre-specified family of comparisons, a correction for multiple comparisons, paired on the same resampled subjects so that shared noise cancels. A difference is reported as confirmed only when this paired test and the sign of the five-fold difference agree, and reported with both otherwise. Because some operations on the GPU are not deterministic, results are reproducible up to seeding rather than bit-for-bit. As a matter of discipline, every confidence interval names the estimator it comes from, and every clinical-utility number names the backbone it is measured on.

3.6.4 Internal evidence, and what would extend it

All of this evidence is internal to a single cohort. Subject-disjoint cross-validation is the strongest guard available within one dataset, and it is not the same as external validation: every split, including the single-split test set, is drawn from the one cohort, not a different population. Confirming that the findings transfer would require a distinct, prospectively collected cohort under a matched protocol, which is the designated next step and is not attempted here.

3.7 The metrics, and why each is used

The clinical motivation for each measure was given in Section 2.2. Here each is defined precisely. At a prevalence near one per cent, almost every heart a screening tool flags will turn out to be healthy. So the metrics that decide whether the tool is useful are the ones read at an operating point a programme could actually run. A single number averaged over thresholds a programme would never use does not decide it. The metrics are organised to match. The clinical-decision metrics lead. The calibration metrics follow as a check on whether the scores can be trusted as probabilities, and the discrimination metrics are reported as supporting evidence, not as the headline.

3.7.1 Sensitivity at a fixed operating point

A screening programme fixes how many healthy pregnancies it can afford to flag and then asks how many affected ones it catches at that setting. The lead operating-point metric is therefore sensitivity at a fixed high specificity, reported at a specificity of 0.95 and again at the stricter 0.99 that is closer to what a national programme would tolerate. The operating point is chosen on each fold as the lowest threshold whose specificity reaches the target, taking the highest sensitivity available there. At a specificity of 0.99 the number of flagged positives per fold is small, so that figure is read as indicative.

3.7.2 What a call means at prevalence

What a clinician experiences is the chance that a flagged heart is truly affected, and that chance depends on how common the condition is. The positive and negative predictive values are therefore reported transported to the deployment prior, not at the enriched prevalence of the research cohort. The transport applies the class-conditional sensitivity and specificity measured on the cohort to a target prior π,

$$\mathrm{PPV}(\pi) = \frac{\mathit{sens} \cdot \pi}{\mathit{sens} \cdot \pi + (1 - \mathit{spec})(1 - \pi)}, \qquad \mathrm{NPV}(\pi) = \frac{\mathit{spec}\,(1 - \pi)}{\mathit{spec}\,(1 - \pi) + (1 - \mathit{sens})\,\pi},$$

evaluated at the population priors of Section 3.2. A predictive value read at the cohort's own prevalence would flatter the tool, because that cohort holds far more disease than a screening population. The transported value is what a deployed test would deliver. The ratio of the positive predictive value to the prior is also reported, as a prior-independent measure of how far a positive call lifts suspicion above the base rate.

3.7.3 Net benefit

Sensitivity and predictive value fix a single operating point. Decision-curve analysis instead asks whether acting on the model beats the two trivial policies of referring everyone and referring no one, across a range of thresholds. The net benefit at a threshold probability p_t is

$$\mathrm{NB}(p_t) = \frac{TP}{n} - \frac{FP}{n}\,\frac{p_t}{1 - p_t},$$

weighing true referrals against false ones by the odds at the threshold, against the references of referring all $(\mathit{prevalence} - (1 - \mathit{prevalence})\,p_t/(1 - p_t))$ and referring none (0) [8]. It is computed within each fold and summarised across folds in the same way as every other metric, and is swept over threshold probabilities of 0.10, 0.15 and 0.20. What the net benefit implies for whether the tool should be used, and the assumption about the relative cost of a missed case that the threshold encodes, are taken up in the discussion. Here the metric is defined and not interpreted.

3.7.4 Calibration

A referral threshold acts on the predicted probability, so it matters whether that probability is meaningful. The Brier score is the mean squared error of the predicted probabilities, a single number that combines calibration and sharpness. Its Murphy decomposition separates out the calibration part as a reliability term. The expected calibration error summarises the gap between confidence and accuracy over ten equal-width bins, a more direct read on calibration alone. Together these say whether the scores can be taken at face value.

3.7.5 Discrimination

The discrimination metrics describe how cleanly the scores separate affected from healthy hearts, irrespective of any threshold. The area under the receiver-operating-characteristic curve is the probability that a randomly chosen affected subject is ranked above a randomly chosen healthy one. It is invariant to any monotone rescaling of the scores and averages over every operating point, including ones a programme would never use. The area under the precision-recall curve is also reported, because a precision-recall summary is sensitive to the rare positive class in a way the receiver-operating-characteristic summary is not. These are reported as supporting checks. What they imply for screening is left to the discussion.

3.8 Reproducibility statement

The evaluation follows one fixed protocol, stated here so that every number in the chapters that follow can be located and reproduced. The analysis plan and the decision rules were specified in advance.

- **Splits.** The canonical protocol is subject-disjoint five-fold cross-validation, subjects assigned to folds by identifier under a fixed global seed. A single stratified 70/15/15 train, validation and test split (same seed, stratified on the binary label) provides a 620-subject single-split cross-check. Cross-validated and single-split numbers are reported separately and never mixed.
- **Estimand.** The per-fold five-fold mean and standard deviation is the sole canonical estimand for every metric, threshold-based ones such as net benefit included. None is pooled across the concatenated predictions of all folds. One estimator and one cohort convention are used throughout: every confidence interval names its estimator and every net-benefit number names its backbone.
- **Intervals.** Confidence intervals on a single metric are subject-level bias-corrected and accelerated bootstraps with one thousand resamples. Paired model comparisons use a subject-clustered bootstrap with two thousand resamples and a correction for multiple comparisons within each pre-specified comparison family.
- **Backbones.** The backbone identities, verified from the extraction code, the embedding dimensionality and each trained head's input dimension, are DINOv2 ViT-B/14 (768-dimensional) as the general-purpose representation and the FetalCLIP ViT-L/14 image tower (projected to 768 dimensions) as the domain-pretrained one.
- **Artefacts.** Every figure and table in the results chapters is regenerated from released code and cached per-subject embeddings under a documented environment. The link to the code is given in the declarations. Results are reproducible up to seeding rather than bit-for-bit, because some GPU operations are non-deterministic.

Step-by-step reproduction instructions, the pinned software environment, and the released per-fold predictions are provided in the project repository (Availability of Data and Materials), so the headline results, figures and tables can be regenerated directly from the released code without access to the raw imaging data.

Chapter 4

Results I: Does It Work

This chapter answers the first of the two questions: whether a light head on frozen foundation-model embeddings separates affected from healthy fetal hearts well enough to be read at an operating point a screening programme could run. It reports discrimination and the sensitivity reached at a fixed high specificity, across the two representations and the three evaluation axes set out in Chapter 3. Whether that separation translates into a useful referral decision at screening prevalence is a separate question, held over to Chapter 5. Unless a number is labelled otherwise, it is a per-fold five-fold mean under the protocol of Section 3.6. Single-split figures, computed on the 620-subject test set, are marked as such and never compared against five-fold numbers.

4.1 Frozen embeddings discriminate, and domain pretraining sharpens the separation

The headline is that the approach works. Table 4.1 reports discrimination for the two heads on all three backbones. The strongest configuration, the FetalCLIP transformer head, reaches an auditable-four AUROC of 0.911 and an all-nine AUROC of 0.891, and at a fixed specificity of 0.95 it recovers 0.593 of affected auditable-four cases. A frozen encoder that was never trained on a heart defect therefore separates the auditable lesions from healthy hearts at a usable operating point. At the stricter specificity of 0.99 the same configuration reaches a sensitivity of 0.314. With only a handful of flagged positives per fold this figure is read as indicative. On the single 620-subject split the sensitivity at 0.95 specificity is 0.435 (single-split), lower than the five-fold mean and reported only as a conservative cross-check.

Holding the head fixed and changing only the representation isolates the effect of domain pretraining. With the transformer head, the domain-pretrained backbone lifts the auditable-four AUROC from 0.825 on the general-purpose backbone to 0.911, a gain of 0.086. With the linear probe the same swap lifts it from 0.818 to 0.871, a gain of 0.053. The two heads differ on the domain backbone, where the transformer separates more sharply than the linear probe (0.911 against 0.871 on the auditable four), and are close on the general-purpose backbone. Discrimination does not settle which head to prefer. That turns on the clinical decision, taken up in Chapter 5.

The separation is well outside the noise. Under the subject-clustered paired bootstrap, the gap between the general-purpose linear probe and the domain transformer head on the auditable four is +0.093 (95% interval $[+0.072, +0.113]$), and the same-size comparison below is significant on the same test.

Table 4.1. Discrimination across the two representations and two heads (per-fold five-fold means). The auditable-four task groups the four referable lesions against healthy controls. The all-nine task is any condition against a structurally normal heart, and sensitivity is reported at a fixed specificity of 0.95. The DINOv2 ViT-L/14 row is the scale-matched control of Section 4.2. Representative per-fold standard deviation is about 0.01 to 0.02, and paired comparisons are in the text.

Representation	Head	All-nine AUROC	Auditable-four AUROC	Sens. at 0.95 spec.
DINOv2 ViT-B/14	mean-pooling linear probe	0.797	0.818	0.355
DINOv2 ViT-B/14	transformer head	0.808	0.825	0.320
DINOv2 ViT-L/14	mean-pooling linear probe	0.824	0.843	0.384
DINOv2 ViT-L/14	transformer head	0.817	0.837	0.326
FetalCLIP ViT-L/14	mean-pooling linear probe	0.855	0.871	0.492
FetalCLIP ViT-L/14	transformer head	**0.891**	**0.911**	**0.593**

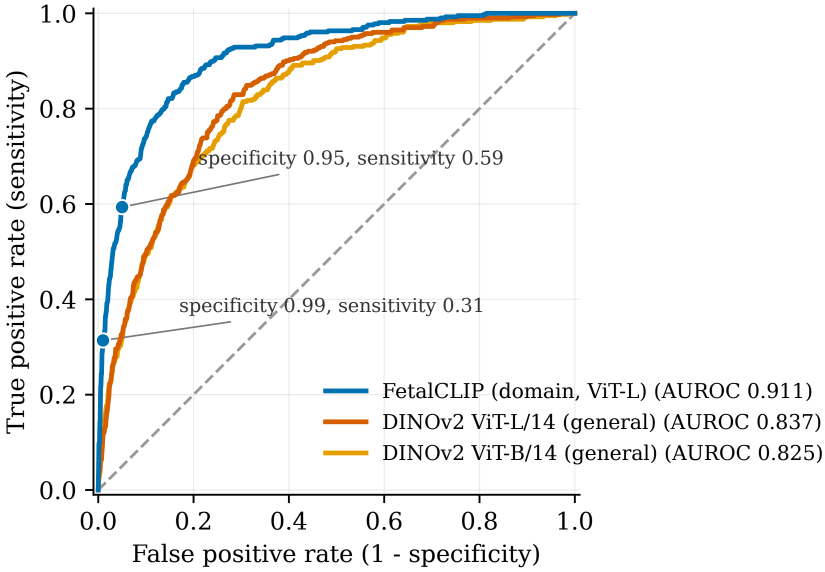

Figure 4.1. Receiver-operating-characteristic curves on the auditable-four task with the transformer head held fixed and only the backbone varied, fold-averaged over the five-fold split. The domain-pretrained FetalCLIP backbone (AUROC 0.911) separates affected from healthy hearts more cleanly than the scale-matched general-purpose DINOv2 ViT-L/14 (0.837) and the smaller DINOv2 ViT-B/14 (0.825). The operating points marked on the FetalCLIP curve recover 59% of cases at 95% specificity and 31% at 99%. The grey dashed line is chance.

4.2 The advantage tracks domain pretraining, not model size

The domain-pretrained backbone is the larger of the two, so a difference in its favour could be a difference of scale. The scale-matched control of Section 3.1 settles this by adding a general-purpose backbone of the same size, DINOv2 at the ViT-L/14 scale. Figure 4.1 draws the transformer-head curve for all three backbones, so both general-purpose sizes are weighed against the domain one on the same axes. Enlarging the general-purpose backbone from ViT-B/14 to ViT-L/14 lifts its transformer-head auditable-four AUROC from 0.825 to 0.837, a gain of only 0.012. At matched ViT-L/14 size the domain-pretrained backbone reaches 0.911 against that general-purpose 0.837, a gap of 0.074 (paired bootstrap $[+0.062, +0.092]$). Of the 0.086 that separates the domain backbone from the smaller general-purpose one, then, scale accounts for 0.012 and the pretraining domain for 0.074, so the discrimination advantage tracks domain pretraining, with most of it left unexplained by model size.

4.3 More of each scan raises discrimination by a modest, steady amount

The third axis varies the per-subject sampling budget with the FetalCLIP transformer head fixed. Figure 4.2 traces discrimination as the budget rises from the baseline of 200 tokens to roughly twenty times that. Over that range the auditable-four AUROC rises from 0.911 to 0.947, a gain of 0.036, and the all-nine AUROC from 0.891 to 0.931. Sensitivity at 0.95 specificity rises further over the same range, from 0.593 to 0.766, a gain of 0.173 or about thirty per cent in relative terms. Spending a fixed budget on more clips rather than more frames within a clip makes little difference. The clip-heavy and frame-heavy points at 400 and 600 tokens differ by 0.007 in AUROC, with an interval through zero, so the frame-against-clip choice is directional only, with the small edge favouring more clips.

Both endpoint gains are significant under the subject-clustered paired bootstrap, the AUROC gain at +0.036 ($[+0.027, +0.047]$) and the sensitivity gain at +0.173 ($[+0.125, +0.243]$). The gains are about one and a half times the unpaired per-fold standard deviation, so the five-fold sign test alone, with five folds, cannot confirm them. They are reported as significant under the paired test and modest in absolute size.

4.4 What the remaining design choices change

The comparisons specified in Table 3.1 form a falsification battery. Each was fixed in advance to rule a candidate mechanism in or out as the binding constraint, so that even a null result is informative: it eliminates that mechanism. Their effect on discrimination is collected in Table 4.2. Each holds the anchor pipeline, the FetalCLIP transformer head at the baseline budget, fixed and changes only the named mechanism. None of them raises discrimination above the anchor. An explicit subject-level hierarchy over the clips lowers the all-nine AUROC from 0.891 to 0.835. Replacing the head with an attention-based multiple-instance aggregator lowers the auditable-four AUROC from 0.911 to 0.840, a drop of 0.071 ($[-0.093, -0.055]$). Dense, contiguous multi-window sampling that captures local motion lowers it by 0.024 ($[-0.038, -0.013]$) on the motion-sensitive lesions. A cardiac-view relevance filter lowers discrimination monotonically as the filter tightens. Re-tuning the training recipe over fifteen configurations moves the all-nine AUROC from 0.873 to 0.877 (single-split, pooled), an interval through zero. Adding head capacity confounds a deeper head with a larger token budget, and for calibration with a change of backbone, so it does not give a clean discrimination reading. It is revisited in Chapter 5. A nine-output multi-label head that predicts each condition directly, instead of slicing the lumped binary head, gives a comparable per-condition macro AUROC (0.885 against 0.893 for the lumped head, a difference of 0.008), with wins traded across conditions. The lumped binary head is retained for simplicity and statistical power. Where that leaves the binding constraint is the subject of the Discussion.

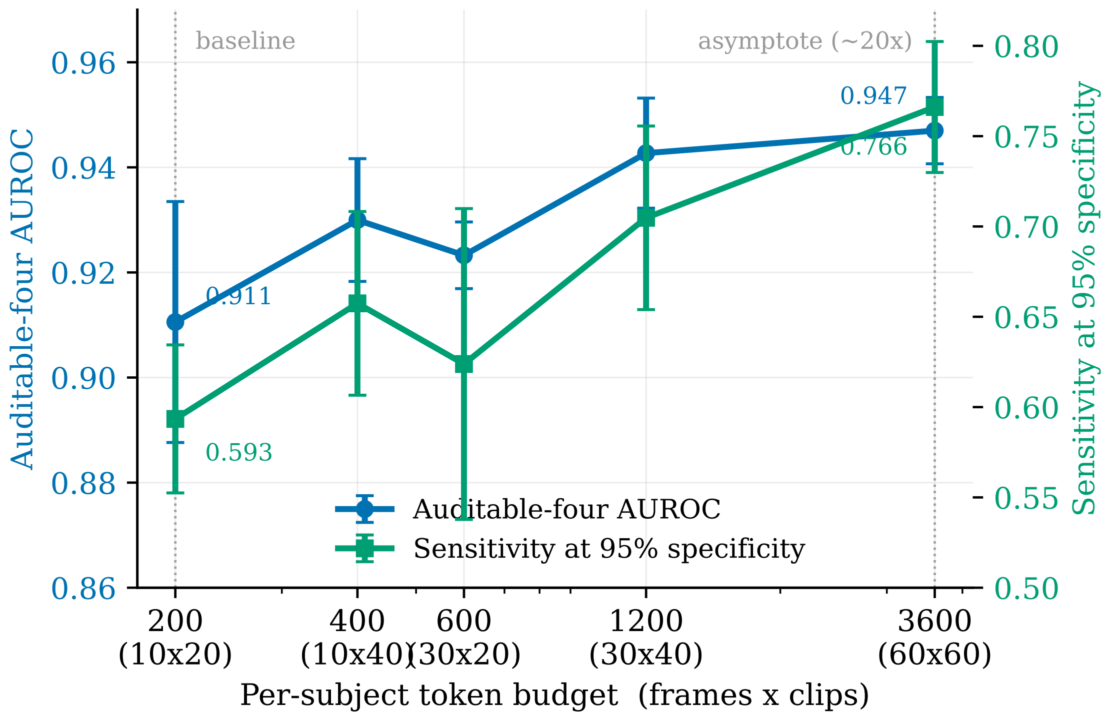

Figure 4.2. The sampling-budget gradient with the FetalCLIP transformer head fixed. As the per-subject token budget grows from 200 to 3600, about twenty times, the auditable-four AUROC rises modestly (0.911 to 0.947) while sensitivity at 0.95 specificity rises more (0.593 to 0.766). Markers are five-fold means and error bars ±1 standard deviation.

4.5 Per-condition discrimination

Read condition by condition (Figure 4.3), discrimination is highest for hypoplastic left-heart syndrome and lowest for the right aortic arch, with the auditable lesions clustered above the rest. Sensitivity at 0.95 specificity for the four auditable lesions (Table 4.3) ranges from 0.527 for transposition of the great arteries to 0.813 for hypoplastic left-heart syndrome, each above the fifty-per-cent floor a screening programme sets itself. With as few as five to six affected subjects per fold for aortic and pulmonary stenosis, their AUROC estimates in Figure 4.3 (the only metric reported for these two lesions) are exploratory. These per-condition figures are read as description: the number of affected subjects is fixed, so no amount of sampling can stabilise them, and the evaluation groups the four lesions into one referable label for its headline.

The postnatal-confirmation cleanup of Section 3.2 acts on the coarctation slice, the condition with the weakest antenatal predictive value. Retraining on the cleaned cohort raises the coarctation slice's five-fold AUROC from 0.858 to 0.879. The cleanup trims the cohort from 4,128 to 4,086 subjects and the coarctation slice from 101 to 59, namely 36 postnatally confirmed cases and 23 that co-occur with another lesion. The gain is small and its interval wide, a labelling-discipline correction made on clinical advice rather than a methodological improvement. Across the sampling budget, the largest per-condition gains fall to the right aortic arch and coarctation, whose AUROC rises by roughly 0.05 from the baseline to the largest budget. With about twenty affected subjects per fold these rises are directional.

Table 4.2. Effect of the specified comparisons on discrimination, against the anchor (FetalCLIP transformer head, 200-token budget). Each row carries its task and estimator, with outcomes only and interpretation deferred to the Discussion.

Change from anchor	Effect on discrimination	Task and estimator	Direction
Flat pooling → subject-level hierarchy	All-nine AUROC 0.891 → 0.835	All-nine, five-fold	lower
Transformer → attention multiple-instance head	Auditable-four AUROC 0.911 → 0.840 (−0.071)	Auditable-four, five-fold	lower
Uniform → dense multi-window sampling	Auditable-four AUROC −0.024 [−0.038, −0.013]	Auditable-four, five-fold (motion subset)	lower
All frames → top-K view filter	Falls monotonically as the filter tightens	Auditable-four, five-fold	lower
Fixed recipe → best of fifteen-configuration search	All-nine AUROC 0.873 → 0.877 (interval through zero)	All-nine, single-split (pooled)	no change
Lumped head → multi-label per-condition head	Per-condition macro AUROC 0.893 → 0.885 (Δ −0.008), wins traded by condition	Per-condition, five-fold	comparable

Table 4.3. Sensitivity at 0.95 specificity for the four auditable lesions (per-fold five-fold, FetalCLIP transformer head), with their cohort positive counts. Per-condition discrimination across all nine conditions is shown in Figure 4.3.

Auditable lesion	Sens. at 0.95 spec.	Affected (cohort)
Hypoplastic left-heart syndrome	0.813	83
Atrioventricular septal defect	0.548	86
Tetralogy of Fallot	0.539	115
Transposition of the great arteries	0.527	128

4.6 Score distributions at the operating point

Figure 4.4 shows where the all-nine binary decision's errors fall. It plots the distribution of the model's predicted probabilities for healthy and for affected subjects, with the threshold that fixes specificity at 0.95 marked. Healthy subjects above the threshold are the false referrals and affected subjects below it are the misses. The two distributions overlap. The affected subjects that fall below the threshold are spread through the healthy range, not bunched just beneath it. The Discussion takes up what that spread means for screening.

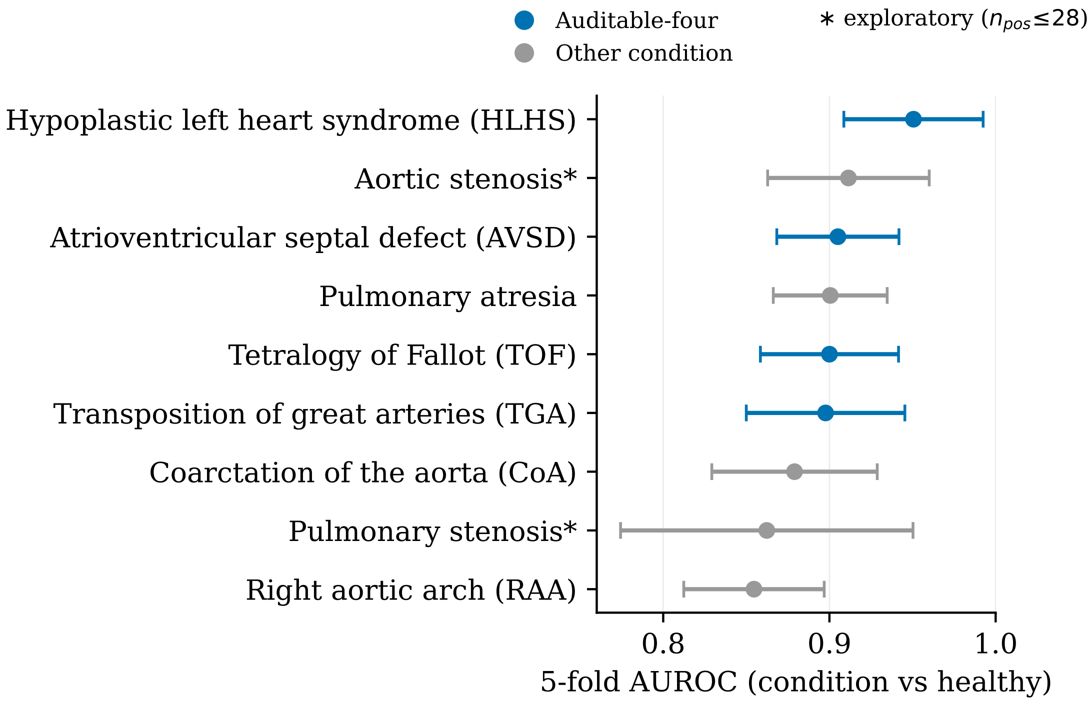

Figure 4.3. Per-condition discrimination of the FetalCLIP transformer head: each marker is the five-fold mean AUROC for detecting that single lesion against healthy controls (one-vs-healthy), with bars at ±1 standard deviation across folds, sorted. AUROC is the probability the model ranks a randomly chosen affected case above a healthy one. The auditable-four lesions are shown in the domain colour and the others in grey. Aortic stenosis and pulmonary stenosis (starred) are exploratory, with at most 28 affected subjects.

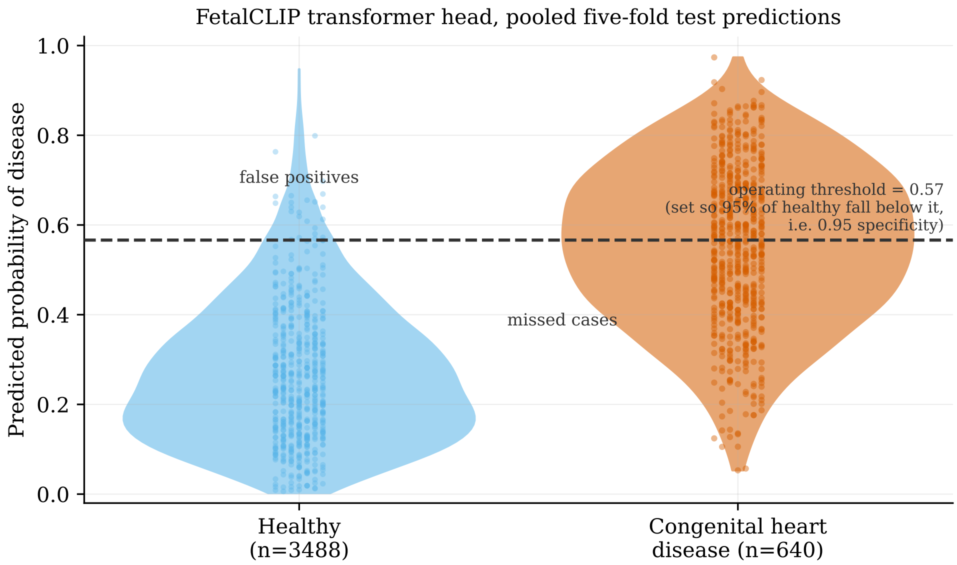

Figure 4.4. Predicted-probability distributions for healthy and affected subjects on the all-nine binary task, pooled across the five folds. The dashed line is the operating threshold set to 0.95 specificity, the score below which 95% of healthy subjects fall. Healthy subjects above it are false positives and affected subjects below it are missed cases. The view is illustrative and pooled across folds. The headline numbers use the per-fold estimand instead.

Chapter 5

Results II: Is It Useful

Discrimination, the subject of Chapter 4, is necessary for screening. On its own it is not enough. This chapter asks the second question: at a prevalence near one per cent, where the only decision is whether to refer, do the embeddings support a useful referral? It reports the metrics a clinician reads, all defined in Section 2.2: the sensitivity reached at a fixed high specificity and what a positive or negative call means at that prevalence, then the net benefit of acting on the model and whether its predicted probabilities can be trusted. As in Chapter 4, every number is a per-fold five-fold mean unless tagged, and every net-benefit figure names the backbone it is measured on. The cost model behind the net benefit, and what these numbers imply for deployment, are taken up in the Discussion.

5.1 Sensitivity at a usable operating point, against the screening baseline

At a fixed specificity of 0.95, meaning about one healthy fetus in twenty is flagged, the best-discriminating configuration, the FetalCLIP transformer head, catches 0.593 of the auditable-four cases (Chapter 4). That figure clears the fifty-per-cent floor a screening programme sets itself [2] and sits above the pooled antenatal detection of congenital heart disease across unselected populations, estimated at about forty-five per cent [3]. Per condition the same operating point catches 0.813 of hypoplastic left-heart syndrome, 0.548 of atrioventricular septal defect, 0.539 of tetralogy of Fallot and 0.527 of transposition of the great arteries. At the stricter specificity of 0.99 the auditable-four sensitivity is 0.314, read as indicative given the small number of flagged positives per fold.

The ceiling is a mature national programme. The United Kingdom audit of the twenty-week scan reports per-condition antenatal detection from roughly seventy to over ninety per cent, for example 0.849 for transposition of the great arteries and 0.927 for hypoplastic left-heart syndrome [9]. The model is below that audit, and the two are not measured at the same operating point: the model is read here at 0.95 specificity while a programme referring at scale operates nearer 0.99. No parity with a trained sonographer is claimed. Against the detection landscape of Figure 2.2 the value the model offers is at the lower-resourced end of the range, where current detection is weakest, not in competition with a strong programme.

5.2 What a positive or negative call means at prevalence

The predictive value of a call depends on how common the condition is, and the cohort is far richer in disease than a screening population. Figure 5.2 transports the positive and

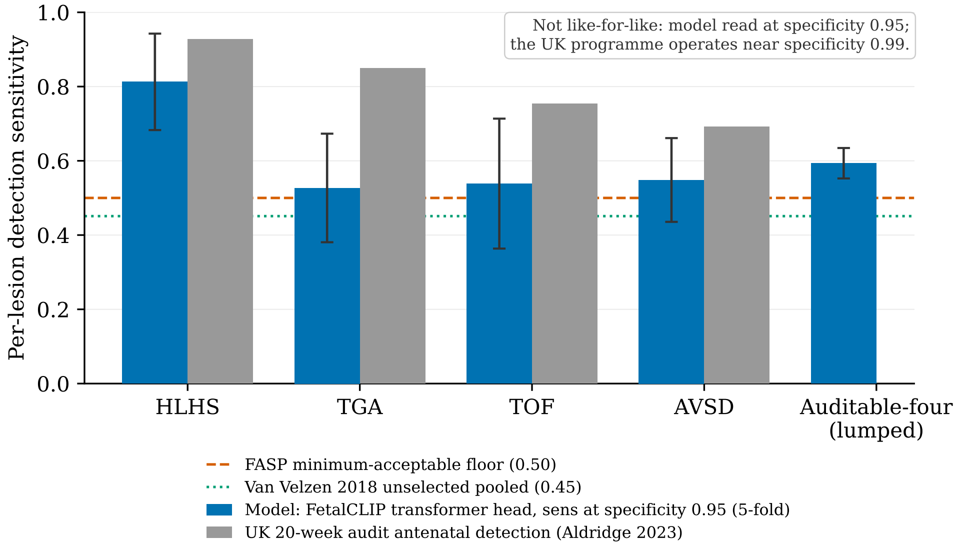

Figure 5.1. Per-lesion antenatal detection: the model (FetalCLIP transformer head at 0.95 specificity, five-fold) against the United Kingdom twenty-week-scan audit [9], with the screening floor [2] and the pooled unselected-population rate [3] marked. The comparison is not like-for-like: the model is read at 0.95 specificity while a national programme operates nearer 0.99. The model clears the floor and the unselected average but sits below the mature programme.

negative predictive values from the cohort to the prevalence a programme would actually see, holding the operating point at 0.95 specificity. The positive predictive value falls from 0.58 at the enriched cohort prevalence, which is not a deployment figure, to about 0.11 at the roughly one-per-cent prior for all congenital heart disease, and to about 0.02 at the four lesions' own population prior of roughly 0.15 per cent. A flag from the model, at deployment prevalence, is therefore far more often a positive in a healthy fetus than a true case. The negative predictive value stays about 0.996 at the one-per-cent prior and higher at lower prevalence. Most of that comes from the base rate, not the model. At a one-per-cent prevalence, reassuring everyone already carries a negative predictive value of 0.99, so the model's 0.996 lowers the residual risk among the reassured from about one in a hundred to about one in two hundred and fifty. That asymmetry is the clinically useful part: at screening prevalence the model is weak at ruling a defect in and strong at ruling one out.

5.3 Net benefit across the decision threshold

A single operating point fixes one trade-off. Decision-curve analysis instead reads net benefit across a whole range of referral thresholds, asking at each whether acting on the model beats referring everyone or no one. Figure 5.3 traces that curve on the auditable-four task across the full clinically-plausible band, and it does not follow the discrimination ranking. Across the low-threshold, miss-averse range a screening service would use, the DINOv2 mean-pooling linear probe stays at or above the refer-none policy. The FetalCLIP transformer head, the stronger ranker, starts there too but falls below refer-none once the threshold passes about 0.13 and stays below it across the rest of the band. At the high-threshold end neither head beats referring no one, a region in which a service would refer almost nobody and which is therefore of little screening interest.

Read at a representative threshold probability of 0.20, naming the backbone in each case, the pattern is concrete. Net benefit is counted in net true referrals per patient, so a value of

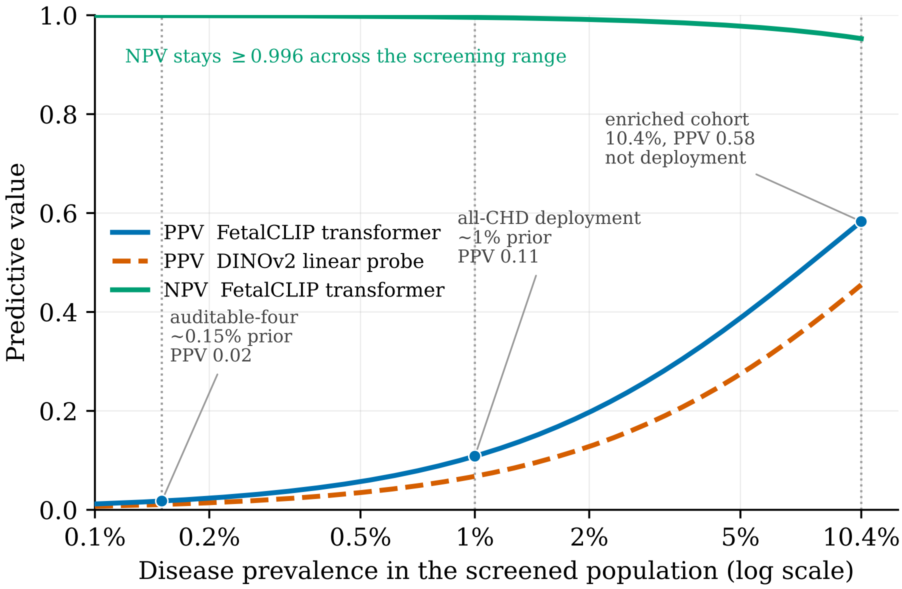

Figure 5.2. Positive and negative predictive value, Bayes-transported across screening prevalence at the auditable-four operating point (specificity 0.95, FetalCLIP transformer head, per-fold five-fold sensitivity and specificity transported to each prior). The positive predictive value falls from 0.58 at the enriched cohort prevalence (10.4 per cent, not a deployment figure) to 0.11 at the roughly one-per-cent all-CHD prior and about 0.02 at the four lesions' own roughly 0.15 per cent prior, while the negative predictive value stays about 0.996 at the one-per-cent prior and higher below it.

+0.01 is about one extra true referral per hundred scans once false referrals are charged at the threshold's cost ratio, and a value below zero means acting does worse than referring no one. The FetalCLIP transformer head returns a net benefit of −0.0372, clearing refer-none on none of the five folds. The mean-pooling linear probe clears it on every fold and on every backbone: +0.0149 on the DINOv2 ViT-B/14 backbone, +0.0250 on the larger DINOv2 ViT-L/14, and +0.0290 on FetalCLIP, each positive on all five folds. On the broader all-nine task the ordering is the same, with the DINOv2 linear probe at +0.0540 (five of five folds) and the FetalCLIP transformer head at +0.0204 (four of five). The best ranker and the best decider are different configurations. The five per-fold curves behind each average are shown in Appendix A. These net-benefit numbers rest on the cost assumption the threshold encodes, which the Discussion takes up along with what the pattern means.

5.4 Whether the predicted probabilities can be trusted

The difference between the heads is one of calibration at the threshold rather than of ranking. The transformer head over-predicts: its mean predicted probability on the auditable-four task is 0.304 against a base rate of 0.104, so at a referral threshold it flags too readily (Figure 5.4), which is what the net benefit penalises. A high area under the ROC curve rewards ordering affected cases above healthy ones, and says nothing about whether a score of 0.3 really stands for a thirty-per-cent chance of disease. The referral decision needs that second thing, the probability right at the threshold, and the two come apart here. By the Brier score, the mean squared error of the predicted probabilities, the transformer head has the lower value (0.110 against the linear probe's 0.142), and a deeper transformer head the lowest of any configuration (0.091). Yet the linear probe, whose predictions sit closer to the base rate where the referral is made, is the one that clears the refer-none baseline.

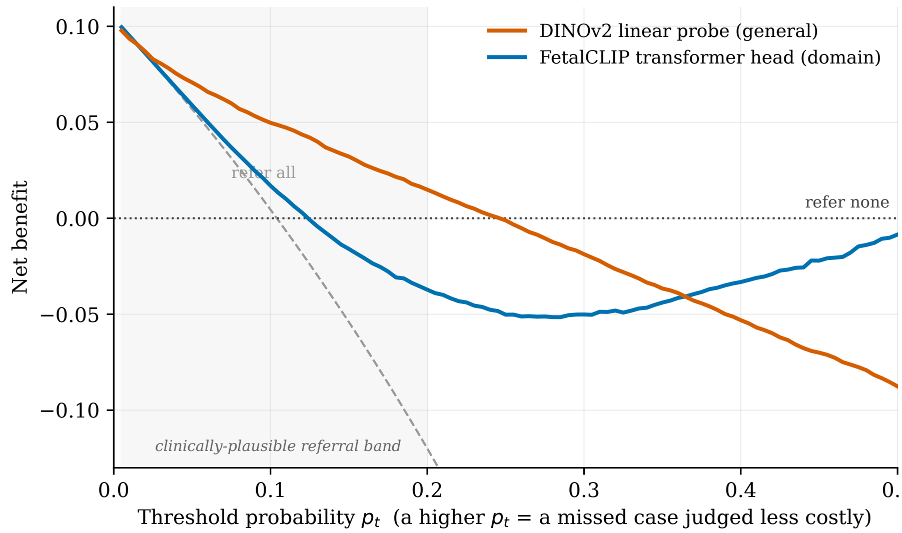

Figure 5.3. Decision-curve analysis on the auditable-four task: five-fold-mean net benefit across the full clinically-plausible threshold band, with the refer-all and refer-none policies as references. The DINOv2 mean-pooling linear probe stays at or above refer-none through the low-threshold, miss-averse range (+0.0149 at a threshold probability of 0.20). The FetalCLIP transformer head, the stronger discriminator, falls below refer-none once the threshold passes about 0.13 (−0.0372 at 0.20). The five per-fold curves are in Appendix A.

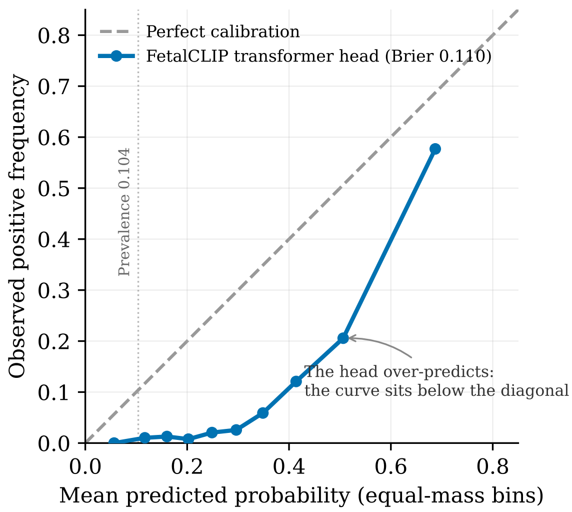

Figure 5.4. Reliability diagram for the FetalCLIP transformer head on the auditable-four task, pooled across folds with equal-mass bins. The head over-predicts: the curve sits below the diagonal, with a mean prediction of 0.304 against a base rate of 0.104 (Brier 0.110). The diagram is pooled across folds, not the per-fold estimand used for the other numbers.

5.5 Training at the task prevalence

A natural explanation for the transformer head's over-prediction is that it was trained with class balancing while the task is rare, so a final comparison trains the FetalCLIP heads at the task's own prevalence instead. It does not reverse the finding. The FetalCLIP transformer head's net benefit improves from −0.0372 to −0.0282 but stays negative, positive on two of the five folds. The FetalCLIP linear probe trained the same way clears the refer-none baseline on all five folds, at +0.041. So the train-and-deploy prevalence mismatch matters, but it is not the whole story. The choice of head continues to decide the referral once the prevalence is matched. What this localises, and what it means for whether such a tool should be used, is taken up in the Discussion.

Chapter 6

Discussion

6.1 Frozen general-purpose embeddings are enough to detect fetal heart disease

Chapter 4 answered the first question, whether the embeddings work: they do, and a domain-pretrained backbone sharpens the separation. Chapter 5 answered the second, whether they are useful. They work as a rule-out test, and the head, not the backbone, is the lever. This chapter reads those two answers against each other.

The central result is one of feasibility. Frozen embeddings from a vision foundation model, trained with no disease label, carry enough cardiac signal for a light head to separate affected from healthy hearts on the iFIND cohort. A model trained only on ordinary images already reaches an auditable-four AUROC of 0.825. Pretraining on fetal ultrasound, still with no disease label, lifts that to 0.911 and catches 0.593 of cases at the 0.95 specificity a screening programme could run. The field has generally assumed that a task this specialised needs a model trained on fetal-cardiac data from scratch. It does not. That a representation learned from natural images already encodes enough of the fetal heart to support the screening decision is the finding this dissertation leads with. It is also what makes the approach cheap enough to be worth pursuing. The expensive component, the backbone, is reused untouched, and only a small classifier is fitted.

Against that backdrop, the benefit of domain pretraining is real but secondary. Replacing the general-purpose backbone with one pretrained on fetal ultrasound lifts discrimination, by 0.086 AUROC on a fixed transformer head, and the scale-matched control attributes that gain to the pretraining domain rather than to model size. At a matched ViT-L/14 scale the domain-pretrained backbone leads the enlarged general-purpose one by 0.074, while merely enlarging the general-purpose backbone recovers only about a seventh of the gap. The real headline is that a general-purpose representation already works, and domain pretraining adds a few points on top of it. For a field weighing whether to invest in collecting and curating a domain corpus, that is the more useful message. The floor set by a general-purpose model is high, and the marginal return of domain pretraining, though genuine, is a refinement at the operating point rather than a different order of capability.

More broadly, the result speaks to access and trajectory. Entry is cheap: a group without the resources to pretrain a model can still build a usable probe by reusing an open backbone and fitting a light head. The approach is also positioned to improve as general-purpose vision models do, since a stronger off-the-shelf backbone can be swapped in and the head refitted. The one direct piece of size evidence here points weakly the same way: enlarging the general-purpose backbone from ViT-B/14 to ViT-L/14 lifted auditable-four AUROC by a small, positive 0.012. That same small number shows scale buys little next to domain pretraining, which bought 0.074 at the matched size. So the in-cohort size signal is too weak to carry a trajectory claim on its own. The trajectory argument rests instead on the field's broader improvement curve for general-purpose models, not on this two-point control, and it is a claim about strategy and economics rather than a measured result of this study.

6.2 Ranking is the wrong headline at one-per-cent prevalence

The discrimination figures above are reported because they are the field's common currency, but they are not the metric a screening decision should be read by. The area under the receiver-operating-characteristic curve averages performance over every threshold, including the permissive ones a programme operating near 0.99 specificity would never adopt, so a model can post a strong AUROC while behaving poorly at the only operating point that matters. This is not to say the measure is empty. There is real signal in it, and a backbone that ranks affected hearts above healthy ones more often is doing something right. The argument is narrower: at a prevalence near one per cent, where the cost of over-referral is what bounds the system, the honest summaries are the sensitivity reached at a fixed high specificity and the net benefit of acting on the model, both read at the operating point a service would actually run. The two results chapters are split along exactly this line for that reason, and the rest of this discussion reads the model by the operating-point metrics, with AUROC kept as a check rather than a headline, as set out in Section 2.2. This is not a theoretical worry. The next section shows the strongest ranker decides worst at the operating point.

6.3 Useful as a rule-out, not yet as a referral test

Whether the model is useful turns on the referral decision, and there the result is two-sided. The best discriminator and the best decider are two different configurations (Figure 6.1). On the auditable-four task the FetalCLIP transformer head, the strongest ranker, returns a negative net benefit at the thresholds a screening programme would use, and clears the refer-none policy on none of the five folds. The simplest head, a mean-pooling linear probe, clears it on every fold and on every backbone. What drives that is calibration at the threshold: the transformer ranks cases well but its scores run too high, so at a referral point it over-refers, while the linear probe's scores sit near the true rate and clear the bar (Section 5.4). This crossover is a secondary finding, below the headline feasibility result. But it matters in practice, because the leaderboard winner and the model worth deploying come apart.

The net benefit that supports this reading rests on an assumption that should be stated plainly. Acting at a threshold probability p_t is to judge a missed defect roughly $(1 - p_t)/p_t$ times as costly as an unnecessary referral: four times as costly at the 0.20 end of the band, and ten or a hundred times as costly towards the low, miss-averse end. Nobody can fix that multiplier for congenital heart disease, where a missed duct-dependent lesion and a false referral do not sit on one scale, and decision-curve analysis, a method from general medicine [8], makes the value judgement explicit without resolving it. The honest response is to read the whole curve rather than commit to a single threshold, and across the clinically-plausible band the ordering of the two heads is robust to wherever a reader sets that cost: the general-purpose linear probe stays at or above refer-none through the miss-averse range, while the stronger-ranking FetalCLIP transformer head falls below it. Reading the finding across the band makes it more honest, not larger. It is still one result

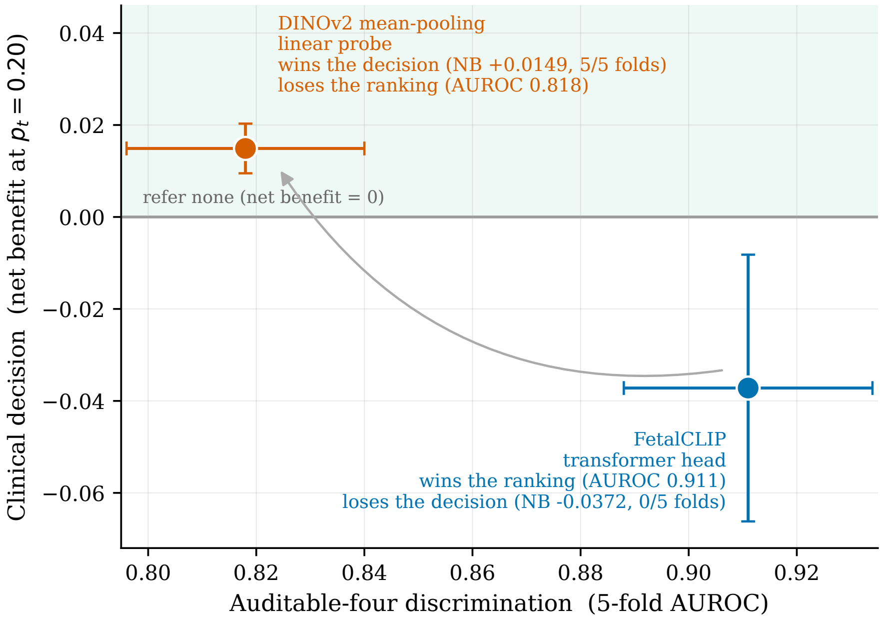

Figure 6.1. The dual finding. Each configuration is placed by its auditable-four discrimination (horizontal) and its clinical decision (vertical, net benefit at a threshold probability of 0.20). The DINOv2 mean-pooling linear probe wins the decision but not the ranking. The FetalCLIP transformer head wins the ranking but not the decision. The shaded region is net benefit above the refer-none policy.

under the second question, with the feasibility result still the lead.

What the predictive values add to the picture is the most clinically legible part. At deployment prevalence a positive call is weak: the positive predictive value falls from 0.58 in the enriched cohort to about 0.11 at a one-per-cent prior and about 0.02 at the four lesions' own prior, so most flags are positives in healthy pregnancies. A negative call is strong: the negative predictive value holds near 0.996. The model is therefore better understood as a rule-out than a rule-in test, a way of reassuring on the large normal majority.

Read against the detection landscape of Section 5.1 (Figure 2.2), that places the tool. It sits above the screening floor and the unselected-population average, but below a mature national programme. Its value therefore lies at the other end of the resource range, where antenatal detection is currently lowest, as a triage or rule-out aid rather than a replacement for expert assessment.

6.4 Where the ceiling sits

The clinical-decision performance does not improve no matter which of the obvious levers is pulled, and that pattern is itself informative. The specified comparisons of Chapter 3 were run precisely to locate the binding constraint, and each one leaves the screening decision where it was. A richer attention-based aggregator lowers discrimination by 0.071, dense motion sampling by 0.024, and an explicit subject-level hierarchy by 0.056. A cardiac-view filter degrades it further as the filter tightens, a fifteen-configuration recipe search moves it within noise, and scaling the per-subject token budget to twenty times the baseline lifts discrimination and sensitivity but leaves the auditable-four net benefit negative at every budget (between -0.037 and -0.079 at $p_t = 0.20$, mean over five folds). Each lever was a candidate explanation for the ceiling. The two aggregation comparisons, an explicit hierarchy and an attention-based aggregator, test whether the weak point is how clips are pooled into a subject decision. Dense motion sampling tests whether the missing signal is temporal. The cardiac-view filter tests whether off-view frames dilute the decision. The token-budget sweep tests whether the head simply needs to see more of each scan. None of them moves the decision, so the constraint is none of them. The most consistent reading is that it is the supervision: a single label per subject, on a few hundred affected subjects, is a weak signal from which to learn a calibrated referral, and no amount of architecture or sampling manufactures information the labels do not carry. Table 6.1 collects these eliminations in one place.

Table 6.1. Where the constraint is not. Every design lever and specified comparison leaves the clinical decision where it was. The best auditable-four net benefit on offer is the DINOv2 linear probe's small +0.0149, and the strongest discriminator never clears the refer-none policy. The pattern locates the binding constraint in the weak subject-level supervision rather than in any component varied.

Lever varied	**Effect on the clinical decision (net benefit at $p_t = 0.20$)**
General-purpose vs domain backbone	Decision unchanged. The strong ranker stays net-benefit-negative on every backbone
Head family	The only positive net benefit is the DINOv2 linear probe's small +0.0149. No head lifts it materially
Sampling budget (200 to 3600 tokens)	Auditable-four net benefit negative at every budget
Subject-level hierarchy	No gain. Discrimination also lower
Multiple-instance aggregator	No gain. Discrimination lower by 0.071
Cardiac-view filter	No gain. Discrimination degrades as the filter tightens
Training-recipe search	No gain, within noise
Added head capacity	Best aggregate calibration, still no decision gain
Training at the task prevalence	Transformer net benefit still negative (two folds of five)

Projected to two dimensions, neither the general-purpose nor the domain-pretrained embeddings separate affected from healthy hearts (Figure 6.2). The silhouette scores are near zero and of opposite sign, +0.018 and −0.011, so neither representation lays the disease out as a distinct cluster. The signal a light head exploits is genuine but high-dimensional and diffuse, and the affected minority does not occupy its own region of the space. That is consistent with a task limited by how much a few hundred subject-level labels can pin down, rather than by whether the embeddings contain the relevant structure at all. The same diffuseness is visible at the operating point: the score distributions of healthy and affected subjects overlap rather than separate, and the misses are spread through the healthy range (Figure 4.4). One plausible systematic false positive is acoustic shadowing across the interventricular septum mimicking a defect, as Chapter 2 described. Pinning that down would need a dedicated error analysis, which the present evaluation flags as future work.

6.5 Relation to prior work

Chapter 2 placed four works around a question none of them answers. The untouched centre was a frozen, general-purpose, medically-unsupervised representation that both

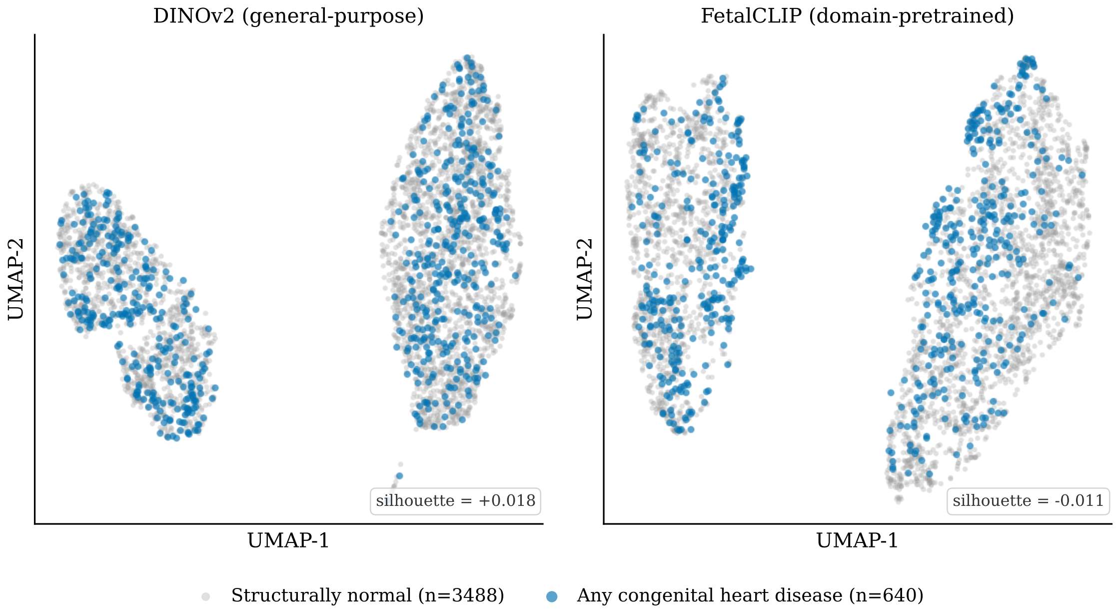

Figure 6.2. Two-dimensional UMAP projections of the per-subject embeddings, coloured by label. Neither the general-purpose nor the domain-pretrained representation separates affected from healthy hearts in two dimensions (silhouette +0.018 and −0.011). The signal a light head exploits is high-dimensional, and the affected minority does not occupy a distinct region.

detects the disease and informs the referral at one-per-cent prevalence, and this work occupies it. What it adds across all four is the same set of levers. It runs a controlled general-against-domain comparison with the head, the data, the splits and the metrics fixed. It adds a scale-matched control that attributes most of the domain gain (0.074 of 0.086 AUROC, at matched ViT-L/14 scale) to pretraining rather than size. It reads clinical utility at screening prevalence. And it builds a falsification ledger that localises the binding constraint to the supervision.

L-FUSION, from the supervisor's own group, supplied the footing, that a frozen encoder's embeddings can be a sufficient statistic for a prediction and its uncertainty [28]. This work pushes that from a domain-pretrained encoder on segmentation to a general-purpose encoder on classification, and turns sufficiency into a measured general-against-domain contrast with a screening-utility readout, though without L-FUSION's formal guarantee or its principled uncertainty. The density model of Schulthess and Konukoglu showed that frozen DINOv2 features stay anatomy-aligned even in abnormal images [27], of which the near-zero silhouettes of Section 6.4 are the fetal-cardiac echo. Their setting is unsupervised normality scoring, where this one is supervised and decision-facing. MM-DINOv2 showed that domain adaptation by fine-tuning works and that its frozen variant was the weakest it compared [29], so frozen probing could not be assumed to succeed. This work demonstrates it on single-stream ultrasound and reads it through a screening lens, without fine-tuning or multiple modalities. STUD is the only prior work on the same target, detecting the disease zero-shot by modelling normality and never weighing the referral at prevalence [30]. The gap this work closes is exactly that last clause, evaluating the detector as a referral test through sensitivity at high specificity, transported predictive values and net benefit. The contribution is therefore positional as well as empirical: each prior work approaches from one side, and this one occupies the untouched centre, bounded by the cost-model caveat below.

6.6 Limitations, and the ethics of the cost model

Several limits bound what these results support. The evaluation is internal to a single research cohort: subject-disjoint folds and a single stratified split used as a cross-check guard against the most obvious leakage, but they are not external validation, and a representation's behaviour can change on a different population, a different scanner fleet or a different case mix. The per-condition figures rest on small numbers, as few as a few dozen affected subjects, well below the samples a clinical prediction model is usually expected to be fitted and validated on [33], so those estimates are read as description and several of their confidence intervals overlap. The operating point is itself a limitation: the model is characterised at 0.95 specificity, and the gap to the roughly 0.99 a national programme runs at is where much of the distance to a mature service lies. And the labels are taken as given, collapsed upstream into nine groups, so any noise or coding choice in them is inherited rather than controlled.

The deepest limitation is in the cost model the utility analysis rests on. Net benefit prices a missed defect against a false referral through a single threshold, but for congenital heart disease the two outcomes are not commensurable in the way that pricing assumes. A missed duct-dependent lesion can be a neonatal death. A false referral is anxiety, cost and a normal echocardiogram. A true early detection can itself lead to a decision about continuing the pregnancy. Reducing these to one number is a value judgement about how a death, a termination, a year of healthy life and a unit of parental anxiety trade off, and that judgement belongs to clinicians, families and policy, not to a decision curve. The net-benefit results here should be read as one quantitative input under one stated assumption. It is not a verdict on whether the tool should be used. The threshold sweep makes the dependence on that assumption visible but does not remove it.

6.7 Future work

The constraint identified above points the way. If the limit is the supervision rather than the representation, the most promising directions add signal on the supervision side: a larger or purpose-built fetal foundation model that brings more of the relevant structure to the surface, and supervision richer than one label per subject, whether clip-level or view-level annotation or weakly-supervised objectives that make better use of the cases that exist. A further open question concerns colour-Doppler imaging. A check found no meaningful imbalance in Doppler use between affected and healthy subjects in this cohort, but the underlying concern is subtler: an operator who already suspects a defect may scan it differently, for instance by adding colour Doppler, so acquisition choices could correlate with disease and be read by the model in place of anatomy. A single dataset cannot fully separate the two, and a study designed to do so would. Three further directions follow from the findings. Because the current approach pools frame embeddings and the temporal comparisons did not recover the motion that mean pooling discards, a video foundation model that natively models the cardiac cycle is a natural next step. A hold-out-disease test, training on a subset of conditions and evaluating detection of a withheld anomaly type, would probe how far the representation generalises beyond the lesions it was fitted on. And an uncertainty estimate that flags low-confidence cases for human review would suit the rule-out role the tool is best placed to fill. The most important next step, though, is external validation: applying the frozen probe unchanged to an independent, prospectively collected cohort under a matched protocol is what would turn a feasibility result on one dataset into evidence that the approach transfers.

Chapter 7

Conclusion

This dissertation tested whether a general-purpose vision foundation model, trained on ordinary images with no medical supervision, already encodes the fetal heart well enough that a light probe on its frozen embeddings can detect congenital heart disease. It can. On the iFIND cohort a frozen general-purpose backbone with only a small trained head separates the auditable lesions from healthy hearts at an auditable-four AUROC of 0.825. A backbone pretrained on fetal ultrasound, still without any disease label, lifts that to 0.911 and reaches a usable screening operating point, 0.593 of cases caught at 0.95 specificity. A task the field has treated as requiring a domain model trained from scratch turns out to be reachable by reusing, untouched, a representation learned elsewhere. That is the result the work leads with, and it lowers the cost of entry to automated fetal-cardiac screening from training a bespoke model to fitting a classifier.

That feasibility carries an operating-point asterisk. At a screening prevalence near one per cent the model works best as a rule-out test. A negative call is strongly reassuring, and a positive call is far more often a healthy fetus than a defect. The best ranker is also a different configuration from the best decider. Read at the operating point a service would run, the tool clears the screening floor and the unselected-population average but sits below a mature national programme, so its place is as a triage or rule-out aid where antenatal detection is currently weakest. It is not a replacement for expert assessment.

The means of reaching these conclusions is itself a contribution. The work is built as a controlled evaluation. It varies one factor at a time across the representation, the head and the sampling budget. It also fixes a single per-fold estimand for every metric, and reads clinical utility by sensitivity at a usable operating point and the prevalence-transported predictive values rather than by ranking alone. The same design turns the comparisons that did not shift the decision into evidence: a battery of specified comparisons, none of which moved the screening decision, locates the binding constraint in the weak subject-level supervision rather than the architecture or the sampling, a finding that would be invisible to a study reporting only its best configuration.

If the limit is the supervision, the way forward is more signal: a larger or purpose-built fetal foundation model, and supervision richer than a single label per subject, validated on an independent prospective cohort. The clinical stakes set out at the start give the result its point: the duct-dependent newborn who collapses days after an apparently normal scan (Figure 1.1) is exactly the case a cheap, reliable rule-out is best placed to catch, and most valuable where antenatal detection is weakest today. That cheapness is durable. Because the backbone is reused frozen, a task can keep pace with improving representations by refitting a light head on a stronger open model rather than pretraining anew. The work shows the embeddings already work, and it shows how to use them well. The general-purpose representations now available off the shelf are stronger than the fetal-imaging field has assumed. A light probe on what they already contain detects fetal heart disease without training a model from scratch.

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Declarations

Use of Generative AI

Generative AI (Anthropic's Claude `https://claude.ai`, via the Claude Code command-line tool) was used during this project for ideation and brainstorming, as reference material that informed my own writing, for generating figures, and for running and cross-checking analyses under configurations I specified. Claude code was also used to write shell scripts for training jobs and all code was fully audited by me. The report is written by me, and all research questions, experimental design, methodological choices, results, interpretations, and conclusions are my own. I confirm that no content generated by Gen AI has been presented as my own work in the report, and I take full guarantee for the correctness of the content.

I also used a data visualisation copilot project that I built (Data Science co-pilot) to help me generate high quality plots from the data from my work. This project is available at `https://abhivir.com/work/data-pilot/`.

I used LLM (Claude: `https://claude.ai`) for help with LaTeX syntax and formatting (table formatting, catching mistakes, displaying figures appropriately).

Ethical Considerations

Data and governance. This dissertation is a retrospective, computational study using anonymised fetal cardiac ultrasound video drawn entirely from existing research datasets acquired through the Imperial College London iFIND research programme, of which the supervisor (Professor Bernhard Kainz) is a co-investigator. No new data were collected and no human-participant intervention or recruitment was carried out for this dissertation. The constituent datasets carry research-ethics-committee approval already in place, including approvals from the London – West London & GTAC Research Ethics Committee (12/LO/1247, 14/LO/1805, 14/LO/1806, 19/LO/1957, 22/LO/0163) and further relevant committees (IRAS 164026, 257568, 292223; REC 20/ES/0005). The use of anonymised patient image data alongside public data has been classified as ethically uncritical by the NHS National Institute for Health Research (NIHR), and data handling follows Information Commissioner's Office (ICO) standards. All experiments reported here are non-invasive and safe: they comprise retrospective analysis of anonymised images and quantitative evaluation of numerical outputs only. Against the Department of Computing ethics checklist, items 6, 7, 26 and 27 apply.

Scope. The outputs are clinical-decision-support research, not a deployed medical device, and any prospective clinical use would require separate regulatory consideration. The clinically asymmetric cost of a missed diagnosis against a false alarm, and the limits of a single-cohort evaluation including the absence of a subgroup-fairness analysis, are discussed in Chapter 6. To the best of the author's knowledge, no further ethical issues remain beyond those covered by the iFIND project's existing approvals.

Sustainability

The computational experiments reported in this dissertation were run on shared GPU resources operated by the Imperial College Department of Computing and on a collaborator-provided workstation. Frozen backbones were used throughout to amortise pretraining cost across all downstream experiments, and per-subject embeddings were cached to disk so that repeated head-training runs incurred no redundant forward passes. Hyperparameter sweeps were budgeted in advance and pruned aggressively when early fold results indicated no signal, and large-batch search was deferred where the incremental compute did not justify the expected gain.

Availability of Data and Materials

All source code, configuration files, experiment manifests, the pinned environment specification (a locked set of dependency versions together with the Python and CUDA versions used), the per-fold model predictions underlying the reported metrics, and the analysis and figure-generation scripts for this dissertation are available in the project repository at https://github.com/abhivir-42/foundation-models-for-fetal-CHD-detection (release tag v1.0-dissertation, the exact snapshot described in this report). The repository README and reproduction guide document the full pipeline end to end: environment setup, embedding extraction, classification-head training, evaluation, and regeneration of every figure and table in this report. They also record the fixed random seed (42), the subject-disjoint five-fold cross-validation rule (fold membership by subject identifier), and the stratified single-split protocol described in Chapter 3. Because the committed per-fold predictions are released, the headline results and all derived figures and tables can be recomputed directly from the repository without access to the raw imaging data, up to small floating-point variation from non-deterministic GPU kernels, which the reproduction guide documents. The underlying iFIND fetal ultrasound corpus is governed by the iFIND research project's data-sharing agreement and is not redistributable. The raw imaging data and the cached per-subject embeddings are therefore not included in the repository, and access requests should be directed to the iFIND principal investigators at Imperial College London.

Separately, the project's complete working repository is available for reference at https://github.com/abhivir-42/fetal-heart-ultrasound-chd. It preserves the full set of experiments carried out during the project, including exploratory analyses and negative results not reported in this dissertation, alongside the extraction, training and evaluation code as it developed. Because this working repository retains intermediate experiment artefacts that reference iFIND subject identifiers, it is held privately under the iFIND data-sharing agreement, with access available to the assessors on request.

Appendix A

Per-fold decision curves

The decision-curve analysis in Section 5.3 reports net benefit averaged across the five cross-validation folds, the canonical estimand used throughout. Figure A.1 shows the five per-fold curves behind that average for the two headline configurations across the full threshold band, so that the spread across folds, not only the mean, is visible. The general-purpose linear probe stays at or above the refer-none policy on every fold across the miss-averse range a screening service would use, while the domain transformer head, the stronger ranker, drops below it on every fold once the threshold passes the low end.

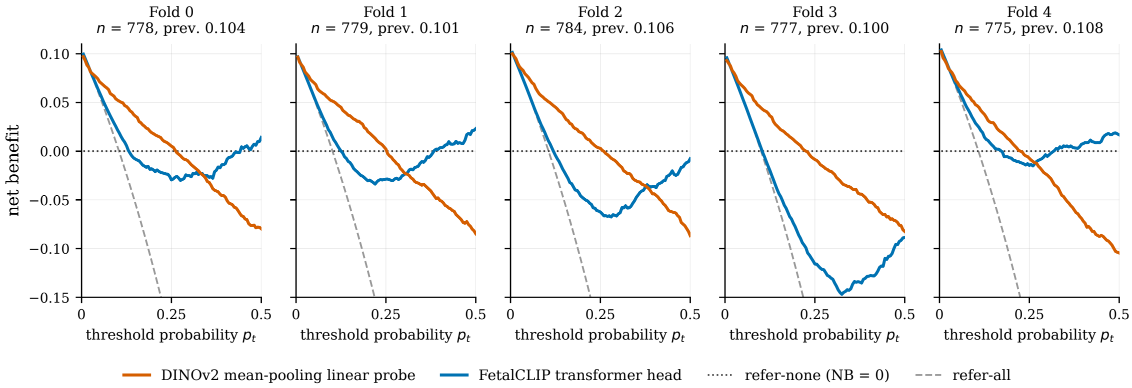

Figure A.1. Per-fold decision curves on the auditable-four task for the DINOv2 mean-pooling linear probe and the FetalCLIP transformer head, the five folds shown individually across the full clinically-plausible threshold band $p_t \in [0, 0.5]$. The averages of these curves are the net-benefit values reported in Section 5.3.